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Master's Degree in Physics of Complex Systems

Master's Thesis

Can a simple model explain economics?

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Abstract

Across datasets, the volatility of a firm's growth rate falls slowly with size, $\sigma(S) \sim S^{-\beta}$ with $\beta \approx 0.15$–0.20, and its growth-rate distribution is fat-tailed. The standard explanation treats each firm as a collection of independently fluctuating sub-units with heavy-tailed sizes, so that both facts arise from the firm's internal composition. Moran, Secchi, and Bouchaud [22] find that firm data contradict this granular picture, and suggest that the shocks hitting a firm's sub-units must become more correlated as it grows. We ask instead whether a dynamical model of interacting firms can generate both on its own.

We model the economy as a Generalized Lotka–Volterra system on a scale-free random graph, in which the interactions act on a firm's size relative to the average firm to make the dynamics scale-invariant. In the strongly disordered regime the economy's composition keeps fluctuating chaotically, and in this persistently fluctuating state the two facts emerge together as stationary properties of the interactions, with no external noise. On the scale-free graph the normalized higher-moment exponents $\zeta_q/\zeta_1 \approx 1$, 1.84, 2.51, 3.02 reproduce the anomalous multiscaling of the empirical 1, 2.0, 2.6, 2.9, and the size–variance relation decreases with firm size, as in the data. Its exponent is set by the graph's degree tail, which brackets the empirical 0.15–0.20: $\beta \approx 0.45$ at tail exponent 2.5, where the ratios sit on the data, falling to $\beta = 0.21$ at exponent 2.2, where it reaches the band while the ratios run a few percent low. The other facts Moran, Secchi, and Bouchaud [22] test against the granular picture, a fat-tailed growth-rate distribution and heavy tails that strengthen with firm size, hold in the model as well.

Contents

1 Introduction **1**

1.1 The granular explanation and where it fails 1

1.1.1 The granular model 1

1.1.2 Where it fails: conditioning on size 2

1.2 A model of interacting firms 4

2 Two traditions **6**

2.1 Compositional and granular firm-growth models 6

2.2 Scale-invariant growth and mean-coupled dynamics 7

2.3 Firm volatility on production networks 7

2.4 Disordered Lotka–Volterra and its fluctuating phase 8

3 The model **9**

3.1 From Lotka–Volterra to the relative model 9

3.1.1 The starting point 9

3.1.2 The relative model 10

3.1.3 The couplings and the graph 11

3.1.4 A convenient change of variables 12

3.2 Phases of the model 13

4 Firm-growth statistics **16**

4.1 Firm-growth statistics in the fluctuating state 16

4.2 Measuring exponents across realizations 17

4.3 Higher moments and the multiscaling 18

4.4 The role of the graph 19

4.5 From degree to size: survival under competition 23

5 Conclusion **25**

A Dynamical mean-field theory of the relative model **30**

A.1 The model and its reduction to a replicator 30

A.2 The local field and the effective single-site process 31

A.3 Self-consistency 32

B Finite-size scaling of the exponents across topologies **33**

C Between-firm growth correlations **36**

D Survival and re-entry dynamics **38**

E Firm-size distribution **40**

F Numerical methods and measurement protocol **41**

F.1 Integration 41

F.2 Volatility estimator 41

F.3 Size–volatility fit 41

F.4 Robustness of the measurement protocol 42

F.5 Code availability 43

During the preparation of this thesis, the author used AI-based tools (Anthropic's Claude) to assist with the development of simulation code and with language editing of the text. All scientific content, including the model design, analysis, and conclusions, is the author's own, and the author takes full responsibility for the content of this work.

Chapter 1

Introduction

Gibrat's law of proportionate effect provides the standard baseline for firm growth. It treats growth over a fixed period as a random shock independent of the firm's current size [16]. A firm's log-size then follows a random walk, and both its average growth rate and its volatility are independent of size. Empirical work has identified two systematic departures from this baseline [26].

The first is in the volatility of the growth rate. Measured year over year, a firm's growth volatility falls with its size as a power law,

$$\sigma(S) \propto S^{-\beta}, \tag{1.1}$$

with a small exponent $\beta \approx 0.15$–0.20 that is stable across datasets [25, 3]. Under Gibrat's law, volatility does not depend on firm size, so $\beta = 0$. At the other extreme, a firm assembled from many independent parts would average their fluctuations down to $\beta = 1/2$. The measured exponent lies between these two benchmarks.

The second concerns the shape of the growth distribution. Independent multiplicative shocks would produce a Gaussian distribution. Instead, the demeaned growth rate is sharply peaked and fat-tailed, close to the Laplace form [25, 6], and retains this shape within every size class. We refer to these two observations as the size–variance law and the growth tent. A model of firm growth must account for both.

1.1 The granular explanation and where it fails

1.1.1 The granular model

The standard explanation treats a firm as an aggregate of many smaller units, each carrying its own idiosyncratic shock. A firm made of K sub-units has size $S = \sum_{i=1}^{K} s_i$, and over a year each sub-unit grows by an independent random shock, $s_i \to s_i(1 + \eta_i)$, with the η_i of mean zero and common variance σ_0^2. The firm's growth rate is then the share-weighted

average of these shocks,

$$g = \frac{\Delta S}{S} = \sum_{i=1}^{K} w_i \, \eta_i, \qquad w_i = \frac{s_i}{S}, \tag{1.2}$$

and, the shocks being independent, its variance is

$$\sigma^2(S) = \sigma_0^2 \sum_{i=1}^{K} w_i^2 = \sigma_0^2 \, H, \tag{1.3}$$

set entirely by the Herfindahl index $H = \sum_i w_i^2$ of the sub-unit shares.

If the units are of comparable size, then $w_i \approx 1/K$ and $H \approx 1/K$, and with K proportional to S we obtain $\sigma(S) \propto S^{-1/2}$, that is $\beta = 1/2$, well above the measured exponent, so the shares cannot be even. H has to decay more slowly than $1/S$, and this happens only when a few large units carry a finite share of the firm and their shocks dominate the sum.

Gabaix [14] established this mechanism at the level of the whole economy, with the units being firms and the aggregate GDP. Firm sizes follow a Zipf-like law [5], a heavy tail $P(s > x) \sim x^{-\nu}$ with ν close to one, and for such a tail the Herfindahl index does not vanish as $1/N$ but much more slowly,

$$H \sim N^{-2(1-1/\nu)}, \qquad 1 < \nu < 2, \tag{1.4}$$

because the sum of squared sizes is carried by the largest few. Their shocks do not cancel, and aggregate fluctuations instead decay as $N^{-(1-1/\nu)}$. Sutton [27] and Wyart and Bouchaud [28] carried the argument back inside a single firm. With $K \propto S$ sub-units whose sizes carry the same heavy tail, $\sigma(S) \propto S^{-(1-1/\nu)}$, so

$$\beta = 1 - \frac{1}{\nu}, \tag{1.5}$$

which lands anywhere between 0 and 1/2 as the tail runs from Zipf ($\nu \to 1$) to thin ($\nu \to 2$). This mechanism explains both observations. The variance falls too slowly with size, and the growth distribution stays fat-tailed because the sum in g is carried by a handful of dominant units. A tail slightly above the Zipf value, $\nu \approx 1.2$, lands the size–variance exponent in the measured range, and on the two facts we began with the granular model works.

1.1.2 Where it fails: conditioning on size

The two observations above are *marginal*: each averages over all firms of a given size, reporting a typical volatility and a pooled growth distribution. Moran, Secchi, and Bouchaud [22] make a stronger test by conditioning on size and examining the full

distribution of growth and volatility within each size class. The granular model makes definite predictions for these conditional distributions, and the data contradict three of them.

First, a firm's growth rate divided by its own volatility is an average of the firm's independent shocks, which by the central limit theorem is closer to Gaussian the more sub-units the firm has, so the largest firms should be the most Gaussian. In the data the rescaled growth stays fat-tailed at every size.

Second, a firm's volatility divided by the typical volatility of firms its size has, in the granular model, a size-independent distribution, which the data confirm. The model, however, also predicts an excess of anomalously volatile large firms, those built from only a few sub-units, whose volatility does not shrink with size; this would appear as a hump in the rescaled distribution, and the data show none.

Third, each higher moment of the volatility distribution falls with size as its own power law, $E[\sigma^q \mid S] \sim S^{-\zeta_q}$, the first exponent being the size–variance law again, $\zeta_1 = \beta$. The granular model predicts the exponents to be independent of the order q. The poorly diversified firms of the previous test keep a volatility of order σ_0 whatever their size, so they contribute σ_0^q times their fraction among firms of size S to the q-th moment. Only that fraction decays with S, and σ_0^q carries no size dependence, so every moment beyond the first is dominated by this contribution and the exponents collapse to a single value, $\zeta_q = \zeta_{q'}$ for all $q, q' \geq 2$. The measured exponents instead keep rising with the order, $\zeta_1, \ldots, \zeta_4 \approx 0.20, 0.39, 0.51, 0.58$ [22]. We call this q-dependence the multiscaling of the size-conditioned volatility: the volatility at a given size is governed not by one scale but by a distribution of them, whose shape changes as the firm grows.

A conditional volatility with a single scale (a fixed distributional shape whose level alone slides with size) gives exponents locked to the first, $\zeta_q = q\,\zeta_1$, a straight line in q, while the granular tail flattens them to a constant beyond $q = 1$. The data lie between these two reference behaviours, rising but sublinearly; Moran, Secchi, and Bouchaud [22] attribute the shortfall below the straight line to subleading corrections from the incipient tail of their volatility distribution. Whatever its asymptotic fate, the measured q-dependence is the observable that separates the data from the granular composition, and it is the one this thesis sets out to generate.

These failures led Moran, Secchi, and Bouchaud [22] to argue that the granular models lack a correlation present in the data: the shocks hitting a firm become "more and more correlated as its size increases, due either to supply chain effects affecting inputs or customers common to all sub-units". Both proposed sources involve relations between firms rather than correlations inside a firm.

1.2 A model of interacting firms

Shared customers and common suppliers are central to the production-network account of the macroeconomy and to the existing network model of firm volatility (Chapter 2). The granular models cannot represent these links: firms are independent, or, in the phrase of Moran, Secchi, and Bouchaud [22], they are "island models". We instead treat firms as the interacting parts and the economy as their aggregate, and ask whether interactions between firms can generate the size-conditioned statistics.

Concretely, we model the economy as a network of interacting firms, each node a company and each edge a signed competitive or cooperative coupling, drawn at random into a disordered interaction matrix. The dynamics are those of the Generalized Lotka–Volterra system from theoretical ecology, each firm growing logistically and coupled to its neighbours through that matrix. Even in the disordered regime where its dynamics stay chaotic, firm sizes in this system remain bounded, so on its own it cannot represent a steadily expanding economy. In the model of this thesis the self-limitation and the interactions instead act on each firm's size relative to the average firm, which makes the dynamics scale-invariant. Scale invariance under a rescaling of all sizes, and the mean-coupled form it takes, are inherited from the econophysics models of wealth and firm sizes (Chapter 2); those models drive the fluctuations with an injected external noise, while here they are generated by the disordered interactions themselves, in a persistently fluctuating state from which we read off the growth statistics of the firms. The model also adds a small immigration term, a firm-entry rate that keeps firms from reaching exact extinction. Total output then grows exponentially without diverging, while the composition of the economy keeps changing chaotically. In this state, the two observations emerge together. Turning off the disorder removes both, which indicates that they are generated by the interactions. The model recovers the anomalous multiscaling of the higher moments and passes the other two size-conditioned tests of Moran, Secchi, and Bouchaud [22]. The size–variance relation is a power law that decreases with size, and its exponent is set by the tail of the graph's degree distribution: changing the tail exponent from 2.5 to 2.2 moves β from 0.45 to 0.21, bracketing the empirical 0.15–0.20. The multiscaling ratios match the data for the 2.5 graph and are a few percent below the data for the 2.2 graph. The relevant feature of the interactions is the degree heterogeneity of the network. Under the standard Lotka–Volterra normalization a firm's noise variance grows with its connectivity, so the graph's degree distribution determines the distribution of firm-level fluctuations. On homogeneous graphs the multiscaling vanishes exactly, and only a scale-free network reproduces the empirical curve. This is how the production-network structure enters the firm-growth statistics.

The rest of the thesis is organized as follows. Chapter 2 places the model in the two bodies of work it draws on, the compositional and scale-invariant models of firm growth on one side and the disordered Lotka–Volterra systems of theoretical ecology on the other, and states what it takes from each. Chapter 3 sets out the model. It introduces the relative

GLV and its scale-invariant growth, says why the obvious GLV will not do and what the relative interaction fixes, and locates the chaotic, growing regime in which the firm-growth statistics are measured. Chapter 4 measures the two facts in the fluctuating state, sets out the estimator that measurement across many realizations requires, recovers the multiscaling and the other two conditional tests that separate the data from the granular models, and then dissects the mechanism: swapping the degree distribution of the graph while holding everything else fixed shows the heterogeneity to be what generates the size-conditioned statistics. Chapter 5 draws the results together and points to what would tie the model more tightly to the data, in particular a direct measurement of the correlations between firms' growth rates that it predicts. Six appendices hold the supporting material: a dynamical mean-field theory of the relative model (Appendix A); finite-size scaling of the exponents across topologies (Appendix B); the correlation analysis (Appendix C); survival and re-entry dynamics (Appendix D); the stationary firm-size distribution (Appendix E); and the numerical methods and measurement protocol (Appendix F).

Chapter 2

Two traditions

The model of this thesis sits at the meeting of two bodies of work that have, for the most part, developed apart. One is the econophysics of firm growth and of the size distributions of firms and wealth: it has the empirical facts and the compositional models built to explain them, and a separate line of scale-invariant growth models that produce heavy-tailed sizes. The other is the statistical physics of large disordered dynamical systems, the random Lotka–Volterra and replicator equations of theoretical ecology, which has the machinery of a persistently fluctuating many-body state and its mean-field theory but has not been turned on firm growth. Neither tradition on its own yields the result of the following chapters. This chapter sets out what each provides, so that the model, and what in it is new, can be read against them.

2.1 Compositional and granular firm-growth models

The granular argument of Chapter 1 says that a firm's growth statistics come from its internal composition: the firm is an aggregate of many smaller units, each fluctuating on its own. Stanley et al. [25] and Amaral et al. [3] first documented the anomalous size–variance scaling. Sutton [27] derived it by treating a firm as a random partition into sub-units, with all partitions equally likely. Wyart and Bouchaud [28] obtained it from sub-units with power-law distributed sizes. Gabaix [14] applies the same logic one level up: when the firm-size distribution is heavy-tailed, the largest firms do not average out, and their idiosyncratic shocks survive as aggregate fluctuations. Bottazzi and Secchi [6] give the distributional companion, a tent-shaped growth distribution generated by cumulative allocation of business opportunities. What these models share, and what Moran, Secchi, and Bouchaud [22] call their "island" character, is that firms do not interact: each firm's statistics come from its own internal composition, and the correlations between firms that a network would carry are absent by construction. Moran, Secchi, and Bouchaud [22] then show that the data break the further predictions of this picture (the thin tail of the rescaled volatility distribution, the persistence of fat tails at all sizes, and the order-dependence of the higher volatility moments) and conjecture a correlation that grows with firm size, of

the kind supply chains and shared customers would produce. A recent working paper by Fontanelli, Napoletano and Secchi [13] reopens even the first fact, reporting a size–volatility relation that steepens for small firms and flattens for large ones rather than holding a single exponent. The compositional tradition thus arrives at the need for a between-firm mechanism it does not itself contain.

2.2 Scale-invariant growth and mean-coupled dynamics

A separate econophysics line builds the scale invariance directly into the dynamics. Bouchaud and Mézard [7] model wealth by

$$\dot{W}_i = \eta_i(t)\, W_i + \sum_j J_{ij} W_j - \sum_j J_{ji} W_i, \tag{2.1}$$

built, in their words, so that "since the unit of money is arbitrary" the dynamics are unchanged when all W_i are multiplied by a common factor. In the mean-field case $J_{ij} = J/N$ this reduces to $\dot{W}_i = \eta_i W_i + J(\bar{W} - W_i)$, each agent exchanging with the population average $\bar{W}$, and the aggregate grows exponentially, $\bar{W}(t) \propto e^{(m+\sigma^2)t}$. The Lotka–Volterra models of wealth and firm market values of the Solomon school [19] likewise couple each unit to the mean, $w_i(t+1) = [1 + \lambda(t)]w_i(t) + a\,\bar{w}(t) - c\,w_i(t)\bar{w}(t)$, though there the coupling is quadratic. Both models produce stationary Pareto or power-law size distributions, consistent with the empirical heavy tail documented by Axtell [5], and in both the fluctuations are supplied by an external multiplicative noise η_i rather than generated by the interactions. The present model inherits its scale invariance and the mean-coupled form of its immigration term from this line, and makes the self-limitation and the disordered interactions relative as well, so that the growth is endogenous and a persistently fluctuating state, rather than a static size distribution, carries the firm-growth statistics.

2.3 Firm volatility on production networks

In production-network models of the macroeconomy, each firm is linked to its suppliers and customers, and a shock to one firm travels along these links. Linked firms fluctuate together, so their shocks do not cancel in the aggregate, and the fluctuations of the whole economy depend on the structure of the network, which is exactly the kind of between-firm structure the granular models leave out [1, 10]. Brought down to the single firm, Herskovic et al. [17] build a customer–supplier network model in which a firm's volatility is set by its position in the network and the concentration of its customer base, and show that the dispersion of the firm-size distribution is the common factor in firm volatilities. This

is the closest existing account to the mechanism of the present thesis, but it is a static, equilibrium model of the *level* of firm volatility, whereas at issue here is the *size-conditioned* structure (the exponent of the size–volatility law and the order-dependence of its higher moments) and its origin in a fluctuating dynamics, which no account in this line addresses.

2.4 Disordered Lotka–Volterra and its fluctuating phase

The dynamics the thesis uses come from a third body of work, the statistical physics of large disordered interacting systems. May's random-interaction argument provides the classical complexity–stability benchmark [21]. The random replicator equation [12, 23] and the disordered generalized Lotka–Volterra model of theoretical ecology [9] describe N units with random pairwise couplings, and a control parameter, the disorder strength σ, sets their collective behaviour. Below a threshold the dynamics relax to a fixed point; above it, related models can exhibit persistently fluctuating states. In a spatially heterogeneous metacommunity these states can be chaotic [24], and a single disordered Lotka–Volterra community with weak immigration displays chaotic turnover [20]. That fluctuating phase is the property the compositional and mean-coupled models above lack and the thesis needs: a state in which every unit keeps fluctuating without external noise, the fluctuations generated by the quenched disorder of the couplings alone [24].

Related analyses use a dynamical mean-field theory (DMFT) that, for the models and assumptions they study, reduces the N-body dynamics to an effective single-unit process in a self-consistent field [15, 24]; Appendix A carries out the corresponding derivation for the relative model. On a heterogeneous graph the reduction becomes degree-resolved: Aguirre-López [2] develops a heterogeneous DMFT in which the field a unit feels carries an amplitude set by its degree, $\sqrt{k_i/C}$, though the analysis is of the fixed points of the non-growing model rather than of the fluctuating phase. In the following chapters, the graph's degree distribution enters the firm-growth statistics through this field. Two features the thesis relies on have close counterparts in recent work on the fluctuating phase: the confinement of abundant units by their own self-limitation and the intermittent turnover of which unit is large [20], and the single-unit process with a migration floor, a finite-correlation-time field, and a self-limitation whose strength grows with abundance [4]. This literature has not carried the fluctuating phase onto a scale-free graph, made the interactions relative so that the aggregate grows, or read off the size-conditioned firm-growth statistics. That is the work of the next two chapters, where the degree heterogeneity turns out to generate the multiscaling the compositional tradition cannot.

Chapter 3

The model

We model the economy as a network of interacting firms whose sizes evolve by Generalized Lotka–Volterra dynamics. We choose these dynamics because they are the least we can assume about a large collection of firms that each grow on their own and interact in pairs, with every firm carrying its own logistic growth and self-limitation and every competitive or cooperative link a single entry of a signed interaction matrix. This is a standard model of a large disordered community in theoretical ecology [9]. The standard form of these dynamics cannot represent a growing economy for two reasons, and this chapter corrects each in turn to arrive at the model we use. The result is then analysed in Section 3.2, where we locate the growing, fluctuating regime in which the firm-growth statistics of Chapter 4 are measured.

3.1 From Lotka–Volterra to the relative model

3.1.1 The starting point

We take as our starting point the Generalized Lotka–Volterra (GLV) system for N firms. Each firm i carries a size $x_i \geq 0$ that grows logistically and is coupled to the others through a signed interaction matrix,

$$\dot{x}_i = x_i \left(1 - x_i + (Wx)_i\right), \tag{3.1}$$

with $W_{ij} = A_{ij}\,\alpha_{ij}$ on a sparse configuration-model graph with a broad, power-law degree distribution. Here A_{ij} is the adjacency matrix of the graph, 1 where firms i and j are connected and 0 otherwise, and C is the mean degree. The coupling strength on each edge is usually taken to be

$$\alpha_{ij} = \frac{\mu}{C} + \frac{\sigma}{\sqrt{C}}\, z_{ij}, \tag{3.2}$$

with the z_{ij} independent standard Gaussian variables, drawn once for each graph and then held fixed, the quenched disorder of the model. The first term is a fixed mean interaction, cooperative for $\mu > 0$ and competitive for $\mu < 0$, and the second is a random part of size σ

that makes every link different. Both are divided by the mean degree so that the pressure a firm feels stays of order one as the graph is made denser, since a firm with about C neighbours picks up a mean of order $C \cdot \mu/C = \mu$ and, summing C independent random terms, a spread of order $\sqrt{C}\,\sigma/\sqrt{C} = \sigma$.

Equation (3.1) fails for two reasons, and the first is that it carries an absolute scale of size that an economy does not have, since the self-limitation $-x_i$ caps an isolated firm at $x_i = 1$ and the quadratic interaction fixes once and for all the size at which growth, competition and self-limitation balance. A real economy has no such reference size. Its output has grown by orders of magnitude over two centuries while the distribution of firm sizes and the statistics of their growth stayed essentially stationary, so the dynamics should be invariant under a rescaling $x \to c\,x$ of the whole economy, which (3.1) is not. The absolute model can be made to grow without bound, but only by tuning the cooperation to the critical strength at which its bounded fixed point disappears, and even there the total abundance diverges in finite time.

The second reason is that even at that point the absolute model delivers only one of the two firm-growth facts we are after, since it reproduces the fat-tailed growth-rate distribution on heterogeneous graphs, but its size–variance relation is not a stationary law, only a transient traced out as an arbitrary initial condition relaxes onto the critical attractor, after which every growth rate vanishes. We need, then, two things the absolute model lacks, invariance under rescaling and a state that keeps fluctuating rather than relaxing.

3.1.2 The relative model

Scale invariance alone fixes the form of the change. For the dynamics to be unchanged under $x \to c\,x$ the right-hand side of (3.1) must be homogeneous of degree one in x, and while the growth term x_i already is, the self-limitation x_i^2 and the interaction $x_i(Wx)_i$ are both of degree two. To bring them back to degree one we divide each by a quantity that scales like a size, and the natural one is the average firm $m = \langle x \rangle$, the typical firm that each one competes against. The chaotic flow also drives some firms to zero size, which is a fixed point of the dynamics, so we add a small immigration term, as migration-supported formulations of disordered ecological models do for the same reason [9, 4]. The constant flux used there would break the scale invariance, so here the entry flow of small firms is set by the size of the economy itself, which prevents exact extinction, and since the term sums to zero over firms it redistributes size without adding to the total. The model of this thesis is then

$$\dot{x}_i \;=\; x_i \left(1 - \frac{x_i}{m} + \frac{(Wx)_i}{m} \right) \;+\; \lambda\,(m - x_i), \qquad m = \frac{1}{N} \sum_j x_j, \tag{3.3}$$

with $W_{ij} = A_{ij}\,\alpha_{ij}$ the disordered couplings (3.2) on the scale-free graph, and $\lambda = 10^{-3}$ the immigration rate. The immigration term reads as entry plus turnover: split as $+\lambda m - \lambda x_i$, it is a constant influx of small firms at rate λm together with a uniform exit of every firm at rate λ, and it sums to zero over the economy, so it reshuffles mass without changing the total. Although it nominally pulls firms above the mean downward, its per-capita effect on a large firm is bounded by $\lambda = 10^{-3}$, uniform across all large firms and three orders of magnitude below the interaction-driven fluctuations, so it cannot tilt the size-conditioned statistics; its effect is confined to small firms, where the influx λm dominates the vanishing multiplicative terms and sets the floor that keeps firms from exact extinction. Its direct dynamical footprint is the small-firm plateau excluded from the fits of Chapter 4; the floor it sets doubles, however, as the survival boundary against which Section 4.5 measures which firms persist. The only changes from the standard system (3.1) are the two factors of $1/m$ and the immigration term, and every term is now homogeneous of degree one: under $x \to c\,x$ the mean m scales the same way, the ratios x_i/m and $(Wx)_i/m$ are unchanged, the immigration flow $\lambda(m - x_i)$ rescales with the economy, and so if $x(t)$ is a solution then so is $c\,x(t)$, leaving the economy with no preferred size. The change is also the economically natural one, since a competitor's pressure on firm i is now set by its market share x_j/m rather than by its absolute size, and a firm's own crowding is measured against the typical firm rather than against a fixed ceiling, so what drives the dynamics is relative standing in the market. The intrinsic Malthusian rate 1 is left untouched and remains the engine of growth, while the interactions, now made relative, only regulate it.

3.1.3 The couplings and the graph

The couplings retain the standard normalization (3.2): every edge is divided by the (realized) mean degree C, the same for every firm, exactly as in the disordered Lotka–Volterra literature. The graph is a configuration model with a power-law degree distribution of exponent $\alpha = 2.5$, built as follows: each firm draws a target degree $k_i = C\,\frac{\alpha-2}{\alpha-1}\,u_i^{-1/(\alpha-1)}$ rounded to the nearest integer, with u_i uniform on $(0, 1)$, the Pareto distribution with tail exponent α and mean C, whose support therefore starts at $k_{\min} = C\,\frac{\alpha-2}{\alpha-1}$, a third of the mean degree at $\alpha = 2.5$, so the ensemble contains no weakly connected firms; the sequence is wired by the configuration model and the graph made simple by deleting self-loops and merging multiple edges [11, 8], and the realized mean degree of the simple graph is the C of the coupling normalization. The graphs of other exponents compared in Chapter 4 change only α in this construction. On such a heterogeneous graph the standard normalization has a consequence on which the results of Chapter 4 will turn: a firm with k_i neighbours sums k_i independent couplings of variance σ^2/C, so the variance of the disorder field it feels grows with its own connectivity, $\mathrm{Var}\,h_i \propto k_i/C$. A firm with more neighbours thus sits in a noisier environment, and the degree distribution of the graph sets the distribution of noise levels across firms. This is one way the topology enters the firm-growth statistics, and a second, the coherent competitive pressure that also grows with a firm's degree,

appears in the survival analysis of Section 4.5.

Under the standard normalization the mean degree must scale with the system, $C \propto N$, the standard Lotka–Volterra normalization whose fully-connected limit is the $C = N$ Gaussian ensemble; we hold the dilution fixed at $C = N/40$. On the power-law graph this keeps the whole degree hierarchy at fixed fractions of N: the largest hub saturates the natural cutoff at $k \approx N$, and the over-coupling of the hubs relative to the mean, of order $k_{\max}/C = N/C$, stays finite as the economy grows. At fixed C, by contrast, the hub over-coupling grows without bound and the size–variance exponent falls toward zero as N grows.

3.1.4 A convenient change of variables

Because the model is scale-invariant, a firm's size matters only relative to the average firm, and the convenient variables are exactly the ones the data use: the scale of the economy, and each firm's size in units of that scale,

$$S_i = \frac{x_i}{m}, \qquad \langle S \rangle = 1, \tag{3.4}$$

where $\langle \cdot \rangle$ denotes the average over firms. The relative sizes carry the composition of the economy, their mean pinned to one by construction, and the average firm m carries its overall scale.

Averaging (3.3) over firms, the immigration term drops (it conserves the total), and

$$\dot{m} = m\, g_{\text{eff}}, \qquad g_{\text{eff}} = \langle Sf \rangle = 1 + \langle S\,(WS) \rangle - \langle S^2 \rangle, \tag{3.5}$$

the size-weighted average fitness of the firms. Averaging the absolute model (3.1) instead leaves a right-hand side quadratic in the scale, the feedback responsible for the finite-time singularity, whereas dividing the interactions by m removes one power of it, so (3.5) is the equation of exponential growth, with no singularity. The rate g_{eff} fluctuates as the composition churns, but its time-average is a small positive constant, so $\log m$ climbs at a steady slope over unbounded time.

While the scale climbs, the composition never settles. From (3.3) and the quotient rule the relative sizes obey

$$\begin{aligned} \dot{S}_i &= S_i\,(f_i - g_{\text{eff}}) + \lambda\,(1 - S_i), \\ f_i &= 1 - S_i + (WS)_i, \end{aligned} \tag{3.6}$$

with the linear fitness f_i and the same size-weighted average $g_{\text{eff}} = \langle Sf \rangle$ as in (3.5), so the economy grows at the average fitness of its firms. This is a replicator equation, the canonical description of frequency-dependent selection in which a firm gains relative size precisely when its fitness exceeds the market average, plus the immigration flow pulling each firm toward the mean size one; it is the standard Lotka–Volterra-to-replicator transformation described by Hofbauer [18]. Note that N appears nowhere in (3.5)–(3.6): the number of

firms enters the model only through the construction of the couplings (the graph and its mean degree $C = N/40$) and as the number of degrees of freedom, not through the form of the dynamics. Because the relative model does not blow up, these equations are written in the same physical time t in which m grows, with no time rescaling. The random-replicator literature identifies the instability of the fixed point above a disorder threshold [12, 23]; in the high-diversity GLV phase the resulting dynamics remain persistently fluctuating and chaotic [24, 20], so the growth rates never vanish, which is the persistently fluctuating state we needed.

The numerical payoff of the split is that we can integrate (3.6) together with $\mathrm{d}\ln m/\mathrm{d}t = g_{\mathrm{eff}}$ directly in physical time, keeping the composition in its bounded variables (the relative sizes have fixed mean one) and the scale in its logarithm, so nothing overflows however large m becomes even though the economy grows by many orders of magnitude over a run; the explicit solver and its cost are specified in Appendix F.

Figure 3.1 shows a single run in this regime. The total abundance (left) climbs as a straight line in its logarithm over the whole integration, the steady exponential growth of (3.5), with no sign of a blow-up. The relative firm sizes $S_i = x_i/m$ (right) do not settle; after an initial transient they keep exchanging rank and spanning orders of magnitude, the persistent chaos of the fluctuating phase. The economy grows while its composition keeps turning over.

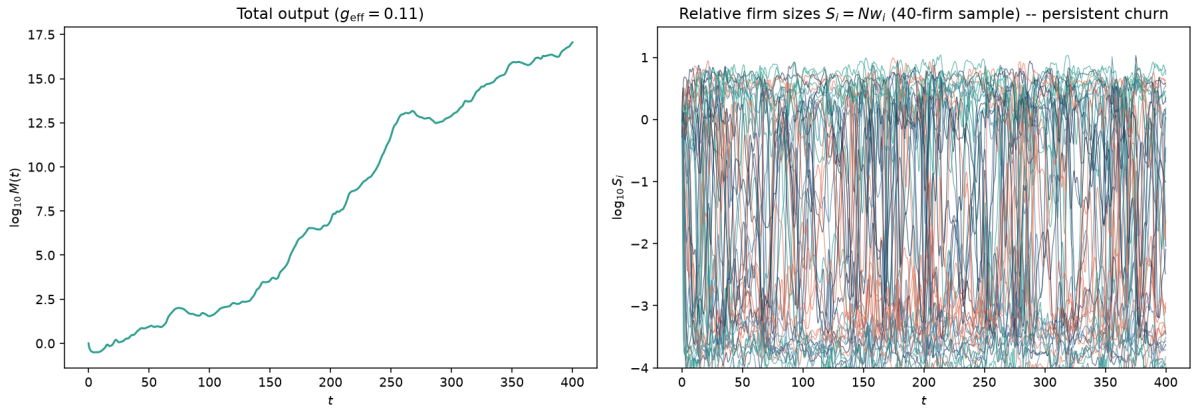


Figure 3.1: A single run of the relative model (3.3) at $\mu = -1.76$, $\sigma = 1.75$, $\lambda = 10^{-3}$ on a power-law graph with $N = 8000$. Left: the total abundance grows exponentially in physical time, its logarithm rising linearly, with no finite-time divergence. Right: a sample of relative firm sizes $S_i = x_i/m$, which keep fluctuating and exchanging rank rather than relaxing to a fixed point.

3.2 Phases of the model

Figure 3.2 maps the behaviour of the model over the plane of the mean interaction μ and the disorder σ, classified from the late-time dynamics of many graph realizations, and two transitions organize it. A dynamical transition separates the frozen region, where the

shape relaxes to a fixed point and the growth rates vanish, from the fluctuating region, where it never settles. Cutting across it, the line $g_{\text{eff}} = 0$ separates a growing economy from a shrinking one. Where the competition is strong and the disorder high, coexistence collapses and the dynamics either condense onto a few firms or grow explosively.

The regime we want is pinned down by three things the data demand, and each one rules out part of the plane. The economy must grow, since output has risen for two centuries, which requires $g_{\text{eff}} > 0$ and discards the whole shrinking half below the $g_{\text{eff}} = 0$ line. Its firms must keep growing period after period, so that firm growth is a stationary statistic, and this requires the fluctuating phase, since in the frozen phase the shape relaxes to a fixed point and every growth rate vanishes, leaving the same transient size–variance relation that already ruled out the absolute model. The economy must also stay diverse, a whole distribution of coexisting firms, which discards the strong-competition, high-disorder corner where coexistence collapses onto a handful of firms or the sizes diverge. What survives all three demands is the chaotic, growing band, where the model reproduces the empirical firm-growth statistics of Moran, Secchi and Bouchaud [22], the region labelled MSB in Figure 3.2. It is the wedge bounded below by the frozen region, on the competitive side by the $g_{\text{eff}} = 0$ line, and on the strong-disorder side by the condensed corner. It is an extended region, so what follows describes the behaviour of the model and not a single tuned operating point.

This phase structure has a mean-field underpinning. The relative model (3.3) is a disordered replicator equation, and its dynamical mean-field theory, worked out in Appendix A, reduces the N-firm dynamics to a single representative firm in a self-consistent Gaussian field. This construction is related to the effective-process treatments of random Lotka–Volterra systems in Galla [15] and Roy et al. [24], although their model-specific assumptions differ from those of the relative model. We locate the freezing threshold directly from simulations: below it the shape relaxes to a fixed point and the growth rates vanish, while above it the flow never settles. The mean interaction enters every firm's fitness as the same uniform shift, so it cancels from the replicator and only sets the level of the growth rate g_{eff}, moving the $g_{\text{eff}} = 0$ line but not the freezing threshold. We locate the threshold and measure the firm-growth statistics directly from the simulations.

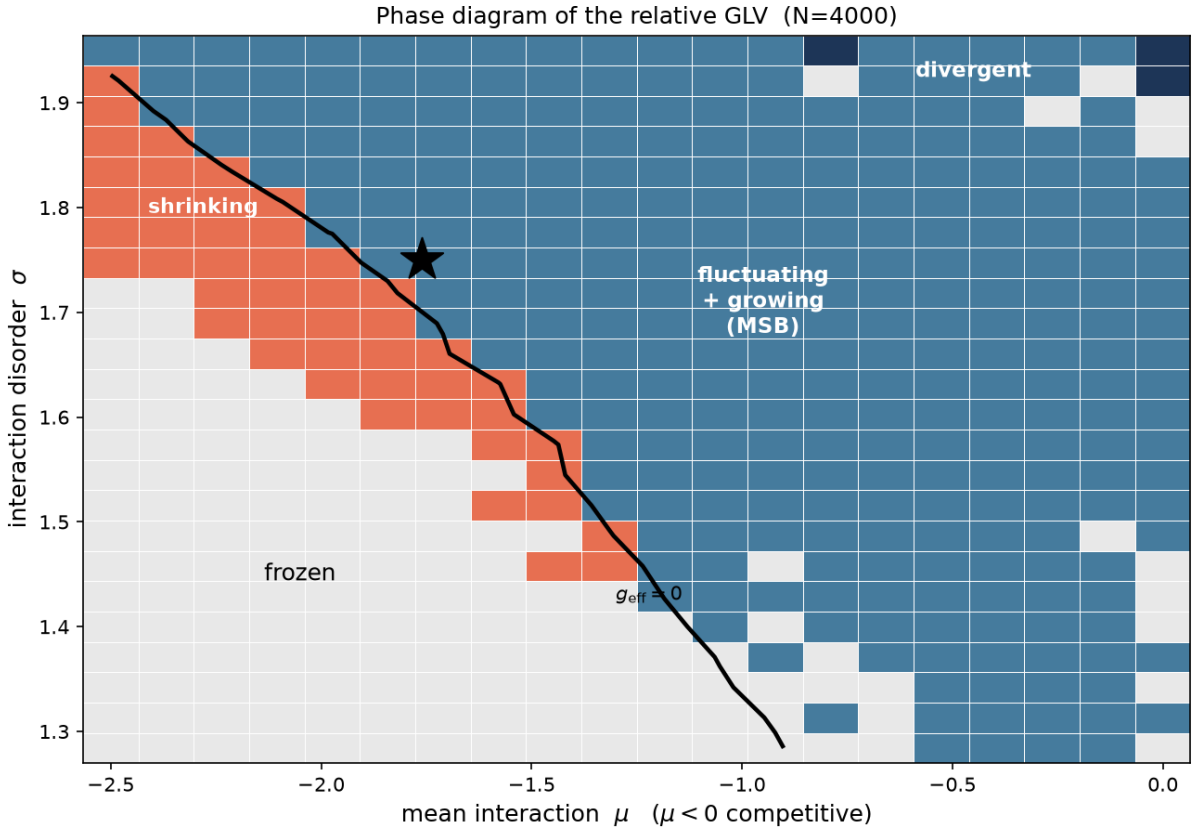


Figure 3.2: Phase diagram of the relative model (3.3), over the mean interaction μ (competitive for $\mu < 0$) and the disorder σ, classified from the late-time dynamics on the power-law configuration-model graph of this section (tail exponent 2.5, mean degree $C = N/40$; $N = 4000$, a 24×20 grid with ten realizations per cell, each drawing its own graph and couplings). At low disorder the shape relaxes and nothing fluctuates (frozen); the freezing threshold rises with the competition strength, from $\sigma \approx 1.3$ at weak competition to $\sigma \approx 1.7$ at $\mu = -2.5$, because the coherent competitive load $\mu\, k_i/C$ grows with degree and a stronger deterministic fitness gradient takes more disorder to destabilize. Above the threshold the dynamics are chaotic, and the solid black line marks the growth boundary $g_{\text{eff}} = 0$, which separates the shrinking region ($g_{\text{eff}} < 0$) from the fluctuating, growing band in which the firm-growth statistics are measured (blue, labelled MSB, for Moran–Secchi–Bouchaud). The band is bounded below by the freezing line, on the competitive side by $g_{\text{eff}} = 0$, and in the weak-competition, strong-disorder corner by the explosive, divergent region. The star marks the operating point $\mu = -1.76$, $\sigma = 1.75$.

Chapter 4

Firm-growth statistics

4.1 Firm-growth statistics in the fluctuating state

We measure the two firm-growth facts on the relative sizes $S_i = x_i/m$ of Section 3.1.4, whose cross-sectional mean is one by construction. The common growth of the economy is divided out in the same step, since $\ln S_i = \ln x_i - \ln m$, so the size and its growth $g_i = \Delta \ln S_i$ are the idiosyncratic part of a firm's trajectory, exactly what the data measure. As in the empirical protocol, each firm contributes while it is listed, that is while S_i stays above the immigration floor, and we read growth rates over a fixed yearly horizon Δt. All statistics are taken in a late window, after the initial relaxation has passed, and restricted to firms that persist across it. They are also taken at $N \geq 8000$: at smaller sizes the fluctuating state is fragile, and at $N = 2000$ nearly half the graph realizations collapse onto a fixed point and freeze, while from $N = 8000$ upward every realization in our ensembles persists.

Two cautions, detailed in Appendix F, shape the measurement. The per-firm volatility is a mean-absolute-deviation estimator robust to the fat tails, and the size–volatility relation is fitted only over its power-law range, the decline above the small-firm plateau set by the immigration floor. The same appendix varies the horizon Δt and the solver tolerance. The moment ratios are insensitive to both; β is stable for horizons up to the correlation time and drifts upward beyond it, and the kurtosis falls with the horizon throughout, so the values quoted below refer to the yearly horizon, as the empirical ones do.

Figure 4.1 shows both observables at the operating point $\mu = -1.76$, $\sigma = 1.75$ marked in Figure 3.2, in the stationary fluctuating state, for twenty independent economies of $N = 16000$ firms each. The size–volatility relation (left) falls monotonically over the fitted range, $\sigma(\bar{S}) \sim \bar{S}^{-\beta}$ with a per-economy exponent $\beta = 0.45 \pm 0.06$ across realizations. Large firms are less volatile than small ones, as in the data. The growth-rate distribution (right) is sharply peaked and fat-tailed, with both flanks heavier than the Laplace reference and no systematic asymmetry between them (Bowley skewness +0.01).

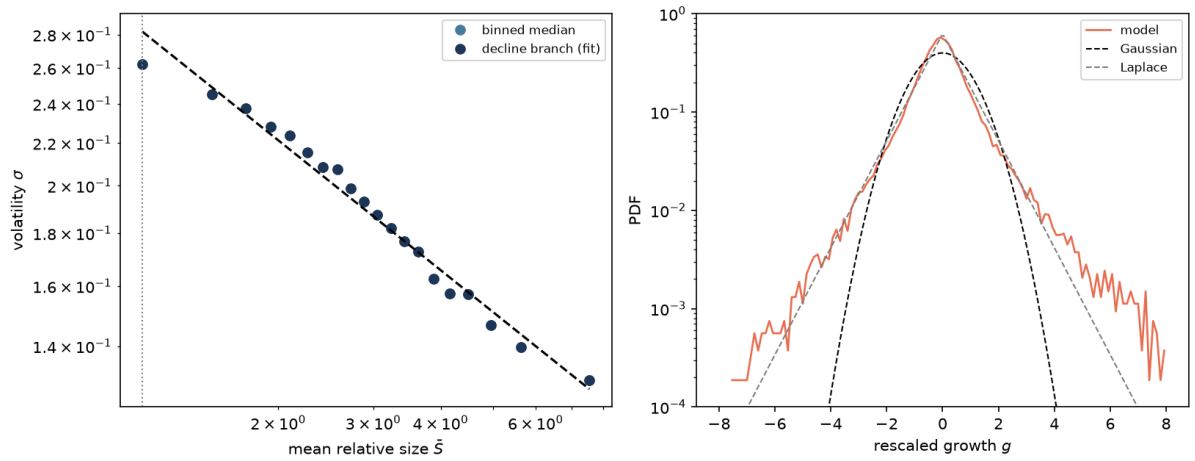


Figure 4.1: Firm-growth statistics at $\mu = -1.76$, $\sigma = 1.75$, pooled over twenty economies of $N = 16000$ firms on the power-law graph ($C = 400$, the fixed dilution $C = N/40$). Left: the size–volatility relation; volatility falls as a power law over the fitted decline above the small-firm immigration-floor plateau, with per-economy exponent $\beta = 0.45 \pm 0.06$ and $R^2 = 0.99$. Right: the growth-rate distribution, sharply peaked and fat-tailed, heavier than the Laplace reference on both flanks and close to symmetric (Bowley ≈ 0), far above the Gaussian.

4.2 Measuring exponents across realizations

Every exponent in this chapter is measured over many graph realizations, and how the realizations are combined affects the result. Each economy carries its own overall volatility scale, set by its particular draw of the couplings, and these scales differ by large factors from one realization to the next. The obvious estimator pools the firms of all economies into one sample and fits the size-conditioned moments once. But conditioning on size across pooled economies then compares a quiet economy's large firms with a loud economy's small ones, which injects size-independent variance into every conditional moment. The injected variance flattens the fitted slopes, more strongly the higher the moment, so the pooled estimator biases every ζ_q downward, bends the ratio curve ζ_q/ζ_1, and the bias does not shrink as more realizations are added: pooling more economies is not more data for this purpose but more mixing.

We therefore use the quenched estimator throughout: each exponent is fitted inside a single economy, on that economy's own decline branch, and the exponents are then averaged over realizations, with the quoted uncertainty the standard error of that average. The two estimators disagree materially. At the operating point of Figure 4.1 the pooled ratios come out 1, 1.77, 2.32, 2.67 while the quenched ratios are 1, 1.84, 2.51, 3.02, a disagreement large enough to reverse a verdict. All claims below are therefore quenched, and the pooled values are reported only for comparison.

4.3 Higher moments and the multiscaling

The size–variance exponent is only the first moment of a richer object, the distribution of growth volatility conditional on size, and it is on the higher moments that the granular models break. Moran, Secchi, and Bouchaud [22] measure the q-th moment of the size-conditioned volatility, $E[\sigma^q \mid S] \sim S^{-\zeta_q}$, and find an exponent that depends on the order q: in the data $\zeta_1, \ldots, \zeta_4 \approx 0.20, 0.39, 0.51, 0.58$. Their granular models predict instead that the three higher moments "decay with size all with the same scaling", a q-independent exponent, and it is this prediction the data contradict. As the introduction set out, two reference lines bound the observable. The granular tail flattens the ratio curve ζ_q/ζ_1 to a constant beyond $q = 1$; a conditional volatility with a single scale (a fixed distributional shape whose level alone slides with size) locks the exponents to the first, $\zeta_q = q\,\zeta_1$, a straight line. The data lie between them, rising with q but sublinearly, and it is this q-dependence, which we call the multiscaling of the size-conditioned volatility, that a model of firm growth must reproduce. (Moran, Secchi, and Bouchaud [22] read the shortfall below the straight line as a subleading correction from the incipient tail of their volatility distribution; whether the empirical sublinearity survives asymptotically is open, and our comparisons are made at the moments and sizes that are measured.)

Figure 4.2 makes the same measurement in the model, the first four moments of the size-conditioned volatility, with each economy fitted on its own decline branch. The quenched exponents rise with the order, $\zeta_1, \ldots, \zeta_4 = 0.41, 0.76, 1.03, 1.24$, with standard errors 0.01, 0.03, 0.04, 0.05 over twenty economies; the first is the size–variance exponent of the previous section. Divided by ζ_1, the q-dependence is 1, 1.84, 2.51, 3.02 (each ± 0.01–0.04), which lies on the empirical 1, 2.0, 2.6, 2.9 through $q = 3$ and slightly above it at $q = 4$, and is separated from the fixed-shape line $\zeta_q/\zeta_1 = q$ by many standard errors at every $q \geq 2$ (right panel). The ratio curve is converged in system size, $\zeta_4/\zeta_1 = 3.09, 3.02, 3.06$ at $N = 8000, 16000, 32000$, and Appendix B carries the finite-size study across all topologies. Two differences in shape remain: the empirical curve saturates hard after $q = 3$ while the model's bends more gently, so the model matches the data's values but not its curvature at the highest moment measured; and the empirical exponents themselves are about half the model's, the magnitude gap studied in Appendix B.

The multiscaling is the first of three size-conditioned tests by which Moran, Secchi, and Bouchaud [22] separate the data from the granular models, and the model passes the other two as well (Fig. 4.3).

The second asks how the spread of firm volatilities depends on size. Divide each firm's volatility by the mean volatility at its size, $\sigma_i/\bar{\sigma}(S)$, and the resulting distribution should be the same at every size, with a bounded tail. The model reproduces this collapse onto a single size-independent shape, with a thin tail (the 99th percentile sits at 2.2 times the median when the rescaling is done per economy), and the collapse tightens, rather than degrading, as the system grows. The granular models predict a second feature here: large firms built from only a few sub-units keep a fixed volatility that does not shrink with

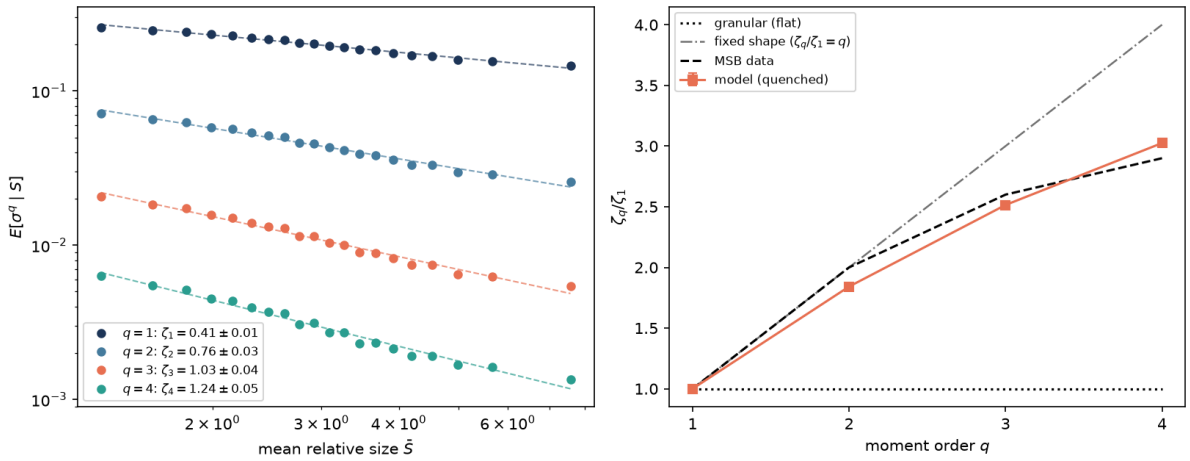


Figure 4.2: Multiscaling of the size-conditioned volatility, $\mu = -1.76$, $\sigma = 1.75$, twenty economies of $N = 16000$. Left: the first four moments $E[\sigma^q \mid S]$ against mean size on a double-logarithmic scale, each fitted on the decline branch; the slopes steepen with the order q, the quenched exponents $\zeta_1, \ldots, \zeta_4 = 0.41$, 0.76, 1.03, 1.24. Right: the exponents normalized by the first, ζ_q/ζ_1, against the moment order, for the model, the data of Moran, Secchi, and Bouchaud [22], the granular prediction of a q-independent exponent, and the fixed-shape line $\zeta_q/\zeta_1 = q$ of a conditional volatility with a single scale. The model bends below the fixed-shape line onto the empirical curve: the shape of the volatility distribution broadens with firm size.

size, which would show up as a hump in the rescaled distribution. Moran, Secchi, and Bouchaud [22] find no such hump in the data, and the model shows none either; it has no sub-units to produce one.

The third test turns on the central limit theorem. If a firm were an aggregate of many independent parts, its growth would look more Gaussian the larger it grew, so the excess kurtosis of its growth would fall toward zero with size. In the data the excess kurtosis does not fall with size; large firms stay at least as fat-tailed as small ones. The model reproduces this: the excess kurtosis of the size-unmixed growth $(g - \bar{g})/\sigma_i$ rises from about two for the smallest firms to about six for the largest, a trend of $+6.5 \pm 0.5$ per decade of size measured economy by economy, positive in every one of the twenty realizations. A firm's growth here is not a sum of independent contributions, so no central-limit averaging thins its tails as it grows.

The model thus clears not only the size–variance exponent that the granular models also fit, but every conditional test by which the data reject them.

4.4 The role of the graph

The couplings put two ingredients into the model, the quenched disorder on the edges and the heterogeneous degrees of the graph that carries them, and the two can be separated. Under the standard normalization (3.2) every edge is divided by the same mean degree, so

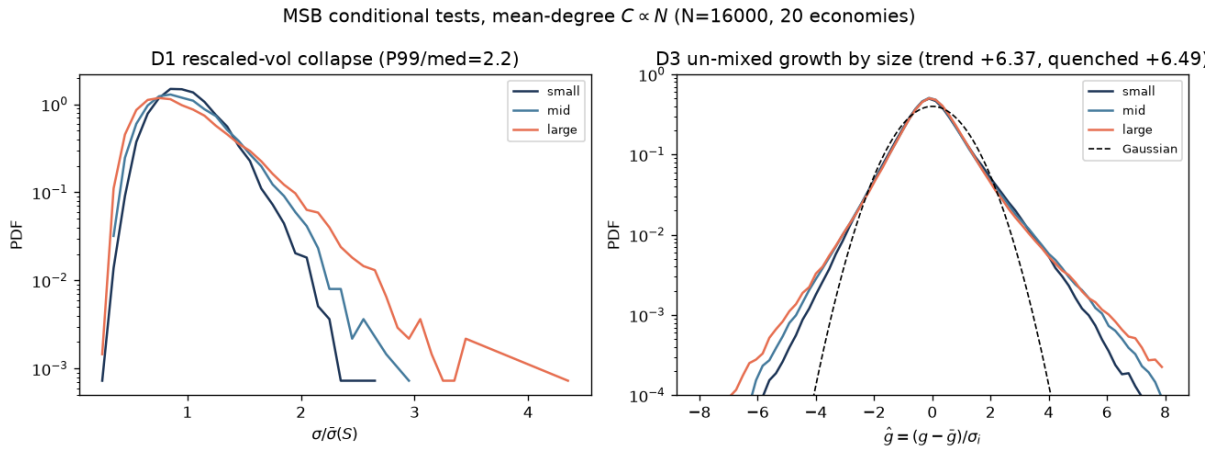


Figure 4.3: The other two discriminating tests of Moran, Secchi, and Bouchaud [22], twenty economies of $N = 16000$. Left (second test): the per-firm volatility distribution, rescaled by its size-conditional mean $\sigma_i/\bar{\sigma}(S)$, collapses onto a single size-independent shape with a bounded tail. Right (third test): the size-unmixed growth $(g - \bar{g})/\sigma_i$ stays fat-tailed at every size, well above the Gaussian, and the larger firms are fatter, the excess kurtosis rising with size in every realization; an aggregate of independent parts would approach a Gaussian.

the variance of the disorder field a firm feels grows with its own degree, $\mathrm{Var}\, h_i \propto k_i/C$, the same degree-dependent field that the heterogeneous mean-field theory of Aguirre-López [2] carries as a factor $\sqrt{k_i/C}$ on its noise, so a firm's degree sets the level of noise it feels. If the multiscaling is really the shape of the volatility distribution broadening with size, this can produce it, and removing the degree heterogeneity should then remove the multiscaling as well.

Figure 4.4 repeats the full measurement with only the degree distribution swapped; the disorder, the operating point and the mean degree $C = N/40$ stay fixed. On the homogeneous graphs the multiscaling vanishes exactly: the fully-connected control gives ratios 1, 2.00, 3.01, 4.00 and the regular graph 1, 1.99, 2.95, 3.87, the fixed-shape line to within error. As the degree distribution broadens the ratio curve bends away from that line, and only the power-law graph with infinite degree variance (exponent 2.5, the graph of every other figure in this chapter) reaches the empirical curve. The size–variance exponent moves with the same parameter, from $\beta \approx 1$ on the homogeneous graphs down through 0.73 (power-law 3.5) and 0.43 (power-law 2.5) to 0.17 (power-law 2.2), inside the empirical band at this size: degree heterogeneity is what pushes the exponent down toward, and through, the empirical range. The tail exponent of the graph is thus a single control parameter that carries both size-conditioned observables across the empirical values: the data are bracketed between exponent 2.2, where β is right and the ratios run slightly low, and 2.5, where the ratios are right and β is high. Pushing the tail harder still (exponent 2.1 and below) overshoots: β falls below the band while half the realizations freeze and the ratio measurement degrades, so the empirical pair sits at the heterogeneous but still stable edge of the family. The growth tent, by contrast, is indifferent to all of it, fat and

symmetric on every topology including the fully-connected one, so the tent belongs to the disorder alone while the size-conditioned structure belongs to the graph.

The power-law-2.2 graph, the bracketing partner that carries the empirical exponent, passes the full battery of conditional tests as well. Run at the same operating point and size as the headline measurements ($N = 16000$, twenty economies, all surviving), it gives $\beta = 0.21 \pm 0.07$, inside the empirical band at a size where Appendix B shows this exponent converged; quenched ratios 1, 1.79, 2.36, 2.72, tracking the empirical curve from below; a rescaled-volatility collapse as tight as the headline graph's (tail P_{99}/median 2.0 against 2.0 at exponent 2.5, with size-independent medians); a size-unmixed kurtosis rising with size at $+5.8 \pm 0.6$ per decade, positive in nineteen of the twenty economies; and a symmetric fat tent (Bowley +0.02, excess kurtosis 10.5). The two bracketing exponents therefore fail in different places and by very different margins: at 2.2 the worst deviation is the highest ratios running 5–7% below the data, while at 2.5 the ratios sit on the data but the exponent overshoots the empirical band by a factor of two and is still drifting. Every conditional test is passed at both.

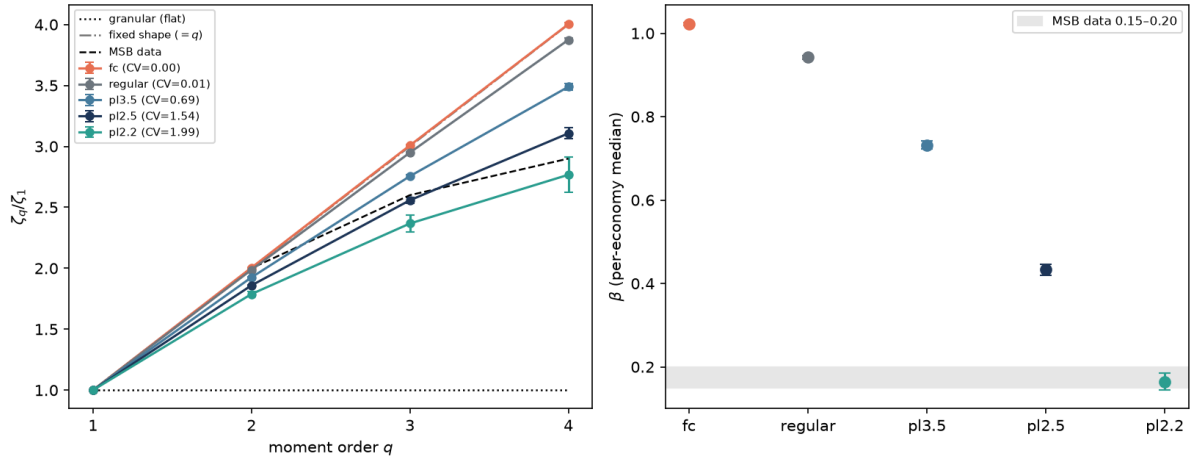


Figure 4.4: The graph generates the size-conditioned statistics. Five degree distributions at the same operating point, mean degree and disorder, twenty economies each ($N = 8000$; the ordering is unchanged at $N = 16000$). Left: quenched multiscaling ratios. The homogeneous graphs (fully connected, regular) sit on the fixed-shape line $\zeta_q/\zeta_1 = q$ (no multiscaling) and the curve bends toward the data as the degree distribution broadens (CV is the coefficient of variation of the degrees); only the infinite-variance power-law graphs reach the neighbourhood of the empirical curve. Right: the size–variance exponent falls from $\beta \approx 1$ on homogeneous graphs toward the empirical range as heterogeneity grows. Table 4.1 gives the numbers.

This ablation is the counterpart of a second one, on the disorder itself. Table 4.2 sweeps σ at fixed μ, λ and N. Below the threshold σ_c the dynamics freeze: the late-time growth fluctuations are zero to the precision quoted, the shape has relaxed to a fixed point, and there is no fluctuating state to measure. The persistent fluctuations and the fat-tailed tent appear only once σ crosses σ_c, growing with the disorder. With no noise

graph	survive	β	ζ_2/ζ_1	ζ_3/ζ_1	ζ_4/ζ_1	tent exkurt
fully connected	20/20	1.02	2.00	3.01	4.00	11
regular	20/20	0.94	1.99	2.95	3.87	11
power law 3.5	20/20	0.73	1.93	2.76	3.49	8
power law 2.5	20/20	0.43	1.86	2.56	3.11	8
power law 2.2	20/20	0.17	1.79	2.37	2.77	9
fixed shape			2	3	4	
MSB data		0.15–0.20	2.0	2.6	2.9	

Table 4.1: The topology sweep of Figure 4.4 in numbers: the same operating point, disorder and mean degree, only the degree distribution swapped ($N = 8000$, twenty realizations per graph). β is the median per-economy decline-branch slope (standard errors 0.003–0.020); the ratios are quenched means over economies (standard errors ≤ 0.05, the power-law-2.2 ratios up to 0.14); the tent excess kurtosis is the per-economy median. Homogeneous graphs sit on the fixed-shape row to within error, the power-law-2.5 graph approaches the empirical ratio row, and the power-law-2.2 graph places β inside the empirical band at this size with ratios slightly below the data; the growth tent stays fat on every topology. (An exponential-degree graph at the same operating point sits at the edge of the freezing transition, losing a third of its realizations, and is omitted.)

in the model, the fluctuations can come only from the disorder, and removing it removes them entirely. The two ablations together assign the phenomena to their sources: the disorder makes the economy fluctuate and gives the growth distribution its heavy tails; the degree heterogeneity of the graph turns those fluctuations into the size-conditioned structure (the low exponent and the multiscaling) that the data show.

σ	growth fluctuation	excess kurtosis	fat-tailed tent?
0.00	0.000	–	no
0.50	0.000	–	no
1.00	0.000	–	no
1.41	0.000	–	no
1.60	0.085	14	yes
1.75	0.175	14	yes

Table 4.2: Disorder ablation at $\mu = -1.76$, $\lambda = 10^{-3}$, $N = 3200$ on a power-law graph. "Growth fluctuation" is the standard deviation of the pooled late-window growth rates. Below the chaotic threshold σ_c the state freezes (growth fluctuation zero to the precision quoted) and there is no genuine fluctuating distribution to characterize, so the kurtosis is left blank; the persistent fluctuations and the fat-tailed tent switch on only once σ crosses σ_c. (A nonzero β can still be fitted below threshold. It reflects a frozen state during its relaxation, the V transient of the absolute critical model, and says nothing about a stationary fluctuating law.)

The interactions-only origin also separates this model from the simplest alternatives. A mean-field multiplicative model with injected noise already gives a growing economy with a stationary heavy-tailed size distribution [7], but there the fluctuations are external and there is no network. Here the same growing aggregate, the heavy tails, and the slow

size–variance decay arise together from the disordered interactions, and the size-conditioned structure from the topology that carries them.

4.5 From degree to size: survival under competition

The degree-dependent field variance that carries the graph into the statistics also acts on the individual firm, and its effect has two parts: degree sets size but barely touches volatility, and it sets size not by advantaging the connected firm but by widening the spread of outcomes among connected firms, most of which lose.

The first part is the observation itself, shown in Fig. 4.5. Measured per economy at the operating point, a firm's mean size rises with its degree, Spearman correlation +0.41 (range +0.37 to +0.44 across ten economies of $N = 16000$), positive in every realization, and the relation is a power law, $\bar{S} \sim k^{\delta}$ with $\delta = 0.56 \pm 0.02$. Unlike the size–variance exponent, δ is converged at our sizes: it is 0.57 ± 0.03 at $N = 8000$ and unchanged at $N = 16000$. The firm's volatility, by contrast, is nearly flat in its degree (Spearman +0.16, right panel), and the flatness is a near-cancellation of two opposing dependences. Fitted jointly across the firms of twelve economies at $N = 8000$, the volatility scales as $\sigma_i \propto k_i^{b}\, \bar{S}_i^{-a}$ with $b = 0.486 \pm 0.012$ and $a = 0.588 \pm 0.015$: at fixed size a hub is noisier, the k_i/C field variance of Section 4.4 surfacing directly, but the size that noise procures quiets the firm through the size–variance law, and the two compose to $\sigma \propto k^{b-a\delta} \approx k^{+0.16}$ with $\delta = 0.56$, the weak residual the figure shows. Degree acts on size, and the volatility inherits only what the size–variance law leaves over.

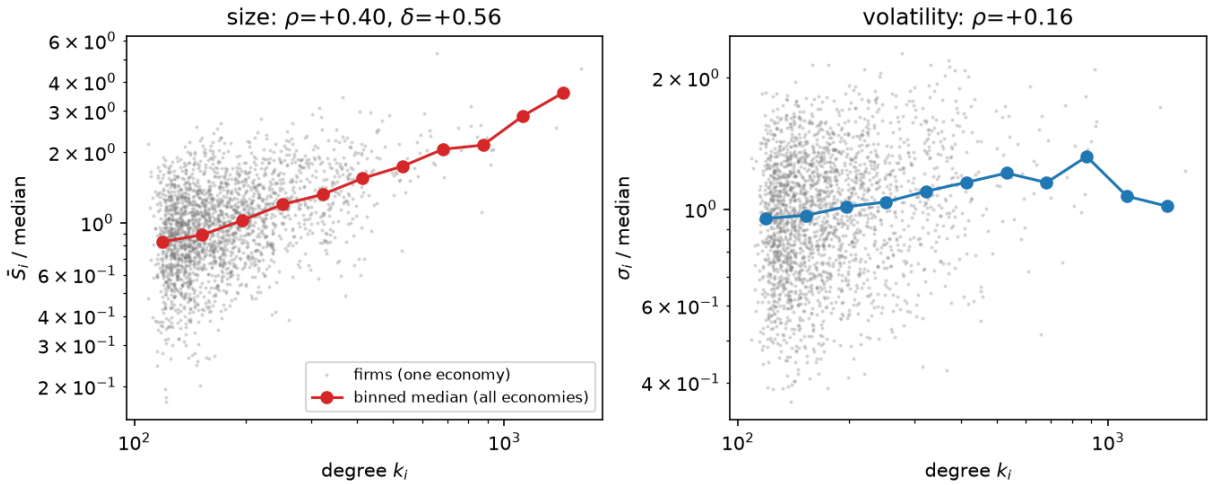


Figure 4.5: Degree and the individual firm, ten economies of $N = 16000$ ($C = 400$), each economy's sizes and volatilities normalized by its own median. Left: mean size against degree; the binned median (across all economies) rises as a power law $\bar{S} \sim k^{\delta}$, $\delta = 0.56 \pm 0.02$, per-economy Spearman correlation +0.41. Right: volatility against degree is nearly flat (Spearman +0.16): the two opposing degree dependences, the noise rising with degree and the size it procures quieting the firm, nearly cancel, and $\sigma \sim k^{+0.16}$ is the residual.

The second part is the selection behind the relation. Under competition ($\mu < 0$) a hub has more competitors and feels a mean suppression that grows with its connectivity, and most hubs indeed lose: the survival probability falls monotonically with degree, from 0.22 ± 0.05 in the lowest degree bin to 0.02 ± 0.02 in the highest. A firm's couplings are also a quenched draw whose spread widens with degree, and conditioning on survival reverses the correlation between the average interaction field a firm sits in and its degree, from -0.27 over all firms to $+0.35$ among the survivors, so the hubs that remain are those whose draw came out favourable, and their sizes track those favourable fields. The field the survivors are drawn from is fat-tailed, a sum over neighbours weighted by their heavy-tailed sizes, and the largest survivors live in exactly that tail.

A final control assigns the diversity itself to the competition. Setting $\mu = 0$ at the same disorder removes the mean suppression, and with it the coexistence: the economy condenses onto a handful of firms (survival below 0.2% in every realization), so the same penalty that eliminates the typical hub holds the rest of the economy in coexistence. Connectivity widens the quenched draw a firm receives, competition removes most of the widened draws, and the surviving tail is large in proportion to the widening. As in Section 4.4, the firm-to-firm structure is quenched, frozen into the realized couplings, and the dynamics select and expose it.

Chapter 5

Conclusion

This thesis asked whether a dynamical model of interacting firms can produce the firm-growth anomalies by itself, without the heavy-tailed internal composition that the granular models assume and that Moran, Secchi, and Bouchaud [22] show the data reject. A Generalized Lotka–Volterra network whose firms are identical, with no injected noise and no built-in correlations, recovers the fat-tailed growth-rate distribution, a size–variance relation that decreases with firm size, as in the data, and every size-conditioned test by which the data reject the granular picture: the anomalous multiscaling of the higher volatility moments, the collapse of the rescaled volatility distribution with a bounded tail, and the persistence of fat tails in the growth of the largest firms. All come out as stationary properties of an economy that grows while its composition keeps fluctuating.

The model is one change to the standard Generalized Lotka–Volterra system. Self-limitation and interactions act on a firm's size relative to the average firm rather than on its absolute size, which makes the dynamics scale-invariant. Total output then grows exponentially, without the finite-time blow-up of the absolute model, and in the strongly disordered regime the composition of the economy never settles but keeps fluctuating chaotically. The couplings use the standard Lotka–Volterra normalization of the disordered literature, on a scale-free graph whose mean degree scales with the number of firms.

The results point to two distinct mechanisms. Switching the disorder off and on shows the fluctuations and the fat-tailed growth distribution to be made by the disordered interactions; they appear on every topology, including a fully connected one. Swapping the degree distribution of the graph, with everything else held fixed, shows the size-conditioned structure (the low size–variance exponent and the multiscaling) to be made by the degree heterogeneity. Under the standard normalization a firm's noise variance grows with its connectivity, so the graph's degree distribution sets the distribution of noise levels across firms. On homogeneous graphs the multiscaling vanishes exactly; as the degree distribution broadens the ratio curve bends toward the data, and only the scale-free graph reaches it. The same heterogeneity reaches the individual firm through survival: most highly connected firms are driven to the immigration floor, and the survivors' sizes grow with degree as $\bar{S} \sim k^{0.56}$, so the degree hierarchy shapes the size hierarchy as well as the noise.

A network account of the *level* of firm volatility, resting on customer concentration, already exists [17]; the present model adds that the *size-conditioned* statistics follow from the heterogeneity of the interaction graph.

Three limitations are important. First, neither the dynamics nor the construction is new, and Chapter 2 places both in their lineage: the fluctuating regime is the chaotic phase of the disordered Lotka–Volterra model [12, 9, 15, 24], and the scale invariance and mean-coupling are those of the econophysics models of wealth and firm sizes [7, 19]. What is specific here is the combination, self-limitation and disordered interactions made relative together, and the finding that the graph's heterogeneity generates the discriminating firm-growth statistics. Second, the quantitative match is partial. The multiscaling ratios and the conditional tests (the shape of the size-conditioned statistics) are reproduced and are converged in system size. The absolute scale is not on the same graph: on the power-law-2.5 graph of the headline figures the size–variance exponent stands at $\beta \approx 0.45$ at $N = 16000$ and 0.49 at $N = 32000$, against the empirical 0.15–0.20, still drifting slowly upward, while on the power-law-2.2 graph the exponent is converged at the upper edge of the empirical band ($\beta \approx 0.21$–0.23) with the multiscaling ratios then running 5–7% below the data; no single tail exponent yet places both on the data at once. The model robustly predicts the shape of the statistics, while their scale remains a finite-size quantity. Third, the growth is near-critical: the operating point sits in the growing band but not far from its boundaries; stronger disorder loses realizations to collapse, and a lighter-tailed graph pushes them into freezing. Perturbing the operating point (σ from 1.5 to 2.0, μ from -1.4 to -2.1) moves the size–variance exponent substantially ($\beta = 0.28$–0.79) while the ratio curve stays bent below the fixed-shape line wherever the fluctuating state survives, so the multiscaling is a property of the phase and the particular exponent values of the operating point.

The exponent's slow upward drift calls for determining its limiting value and for developing an analytical theory. The mean-field treatment of Appendix A stops at the relaxed phase on the homogeneous graph. The size distribution, measured in Appendix E, is effectively truncated where the data are Zipf-like, so matching it will take a modified model, not a larger simulation. The immigration floor's plateau on the smallest firms is worth a second look, now that Fontanelli, Napoletano, and Secchi [13] report an empirical size–volatility relation that steepens for small firms and flattens for large ones. The correlation structure is the sharper question. Moran, Secchi, and Bouchaud [22] conjecture that the granular models fail for lack of a between-firm growth correlation that grows with size; Appendix C measures that correlation directly and finds it modest and roughly independent of both size and degree. The model therefore reproduces the size-conditioned statistics through the degree-dependence of the noise amplitude, not through the size-dependent correlations proposed by Moran et al. Whether real firms carry the size-growing or the flat correlation is a question for firm-level data on a known network. Incorporating couplings from real input–output and supply linkages would connect the model more directly to the macroeconomic dynamics behind these stylized facts.

Bibliography

[1] Daron Acemoglu et al. "The Network Origins of Aggregate Fluctuations". In: *Econometrica* 80.5 (2012), pp. 1977–2016. DOI: 10.3982/ECTA9623.

[2] Fabián Aguirre-López. "Heterogeneous mean-field analysis of the generalized Lotka–Volterra model on a network". In: *Journal of Physics A: Mathematical and Theoretical* 57.34 (2024), p. 345002. DOI: 10.1088/1751-8121/ad6ab2.

[3] Luís A. N. Amaral et al. "Scaling Behavior in Economics: I. Empirical Results for Company Growth". In: *Journal de Physique I* 7.4 (1997), pp. 621–633. DOI: 10.1051/jp1:1997180.

[4] Théo Arnoulx de Pirey and Guy Bunin. "Many-Species Ecological Fluctuations as a Jump Process from the Brink of Extinction". In: *Physical Review X* 14.1 (2024), p. 011037. DOI: 10.1103/PhysRevX.14.011037.

[5] Robert L. Axtell. "Zipf Distribution of U.S. Firm Sizes". In: *Science* 293.5536 (2001), pp. 1818–1820. DOI: 10.1126/science.1062081.

[6] Giulio Bottazzi and Angelo Secchi. "Explaining the Distribution of Firm Growth Rates". In: *The RAND Journal of Economics* 37.2 (2006), pp. 235–256. DOI: 10.1111/j.1756-2171.2006.tb00014.x.

[7] Jean-Philippe Bouchaud and Marc Mézard. "Wealth condensation in a simple model of economy". In: *Physica A: Statistical Mechanics and its Applications* 282.3–4 (2000), pp. 536–545. DOI: 10.1016/S0378-4371(00)00205-3.

[8] Tom Britton, Maria Deijfen, and Anders Martin-Löf. "Generating simple random graphs with prescribed degree distribution". In: *Journal of Statistical Physics* 124 (2006), pp. 1377–1397. DOI: 10.1007/s10955-006-9168-x.

[9] Guy Bunin. "Ecological communities with Lotka–Volterra dynamics". In: *Physical Review E* 95.4 (2017), p. 042414. DOI: 10.1103/PhysRevE.95.042414.

[10] Vasco M. Carvalho and Alireza Tahbaz-Salehi. "Production Networks: A Primer". In: *Annual Review of Economics* 11 (2019), pp. 635–663. DOI: 10.1146/annurev-economics-080218-030212.

[11] Michele Catanzaro, Marián Boguñá, and Romualdo Pastor-Satorras. "Generation of uncorrelated random scale-free networks". In: *Physical Review E* 71 (2005), p. 027103. DOI: 10.1103/PhysRevE.71.027103.

[12] S. Diederich and M. Opper. "Replicators with random interactions: A solvable model". In: *Physical Review A* 39.8 (1989), pp. 4333–4336. DOI: 10.1103/PhysRevA.39.4333.

[13] Luca Fontanelli, Mauro Napoletano, and Angelo Secchi. *Rethinking volatility scaling in firm growth*. LEM Working Paper Series 2025/27. Laboratory of Economics and Management (LEM), Sant'Anna School of Advanced Studies, 2025. DOI: 10.2139/ssrn.5399682.

[14] Xavier Gabaix. "The Granular Origins of Aggregate Fluctuations". In: *Econometrica* 79.3 (2011), pp. 733–772. DOI: 10.3982/ECTA8769.

[15] Tobias Galla. "Dynamically evolved community size and stability of random Lotka–Volterra ecosystems". In: *Europhysics Letters* 123.4 (2018), p. 48004. DOI: 10.1209/0295-5075/123/48004.

[16] Robert Gibrat. *Les Inégalités Économiques*. Paris: Recueil Sirey, 1931. URL: https://ci.nii.ac.jp/ncid/BA3871466X.

[17] Bernard Herskovic et al. "Firm Volatility in Granular Networks". In: *Journal of Political Economy* 128.11 (2020), pp. 4097–4162. DOI: 10.1086/710345.

[18] Josef Hofbauer. "On the occurrence of limit cycles in the Volterra-Lotka equation". In: *Nonlinear Analysis: Theory, Methods, Applications* 5.9 (1981), pp. 1003–1007. ISSN: 0362-546X. DOI: 10.1016/0362-546X(81)90059-6. URL: https://www.sciencedirect.com/science/article/pii/0362546X81900596.

[19] Ofer Malcai et al. "Theoretical analysis and simulations of the generalized Lotka–Volterra model". In: *Physical Review E* 66.3 (2002), p. 031102. DOI: 10.1103/PhysRevE.66.031102.

[20] Emil Mallmin, Arne Traulsen, and Silvia De Monte. "Chaotic turnover of rare and abundant species in a strongly interacting model community". In: *Proceedings of the National Academy of Sciences* 121.11 (2024), e2312822121. DOI: 10.1073/pnas.2312822121.

[21] Robert M. May. "Will a Large Complex System be Stable?" In: *Nature* 238.5364 (1972), pp. 413–414. DOI: 10.1038/238413a0.

[22] José Moran, Angelo Secchi, and Jean-Philippe Bouchaud. "Revisiting Granular Models of Firm Growth". In: *arXiv preprint arXiv:2404.15226* (2024). DOI: 10.48550/arXiv.2404.15226. arXiv: 2404.15226 [econ.GN].

[23] M. Opper and S. Diederich. "Phase Transition and 1/f Noise in a Game Dynamical Model". In: *Physical Review Letters* 69.10 (1992), pp. 1616–1619. DOI: 10.1103/PhysRevLett.69.1616.

[24] Felix Roy et al. "Complex interactions can create persistent fluctuations in high-diversity ecosystems". In: *PLoS Computational Biology* 16.5 (2020), e1007827. DOI: 10.1371/journal.pcbi.1007827.

[25] Michael H. R. Stanley et al. “Scaling behaviour in the growth of companies”. In: *Nature* 379.6568 (1996), pp. 804–806. DOI: 10.1038/379804a0.

[26] John Sutton. “Gibrat’s Legacy”. In: *Journal of Economic Literature* 35.1 (1997), pp. 40–59. URL: https://EconPapers.repec.org/RePEc:aea:jeclit:v:35:y:1997:i:1:p:40-59.

[27] John Sutton. “The variance of firm growth rates: the ‘scaling’ puzzle”. In: *Physica A: Statistical Mechanics and its Applications* 312.3–4 (2002), pp. 577–590. DOI: 10.1016/S0378-4371(02)00852-X.

[28] Matthieu Wyart and Jean-Philippe Bouchaud. “Statistical models for company growth”. In: *Physica A: Statistical Mechanics and its Applications* 326.1–2 (2003), pp. 241–255. DOI: 10.1016/S0378-4371(03)00267-X.

Appendix A

Dynamical mean-field theory of the relative model

This appendix derives a dynamical mean-field theory (DMFT) for the scale-invariant model of the growing-economy chapter, where interactions act on the average firm rather than on absolute abundances. The model reduces to a disordered replicator equation. Its DMFT is therefore the heterogeneous single-site process used to locate μ_c by Aguirre-López [2], with two additions required by scale invariance. We derive the reduction, the effective single-site process, and the associated self-consistency conditions.

We set $\lambda = 0$ here; the numerical model retains $\lambda = 10^{-3}$ to prevent extinction, whose small mass-conserving effect is neglected in the bulk DMFT.

A.1 The model and its reduction to a replicator

We take the relative GLV in the competition convention used in the simulations,

$$\dot{x}_i = x_i \left[1 - \frac{x_i}{m} - \frac{(\alpha x)_i}{m} \right], \qquad m = \frac{1}{N} \sum_j x_j = \frac{M}{N}, \tag{A.1}$$

with couplings on a graph of mean degree C,

$$\alpha_{ij} = A_{ij} \Big(- \frac{\mu}{C} + \frac{\sigma}{\sqrt{C}}\, z_{ij} \Big), \qquad z_{ij} \sim \mathcal{N}(0,1), \qquad \overline{z_{ij} z_{ji}} = \gamma, \qquad \alpha_{ii} = 0. \tag{A.2}$$

This layout differs from (3.3) and (3.2) by two compensating sign flips (the interaction is subtracted and the mean coupling written $-\mu/C$), so μ is the same parameter in both, competitive at the operating point $\mu = -1.76$. Both the self-limitation x_i/m and the interaction $(\alpha x)_i/m$ are divided by the population mean m. The right-hand side of (A.1) is then homogeneous of degree one in x, so the bracket $F_i := 1 - x_i/m - (\alpha x)_i/m$ depends on x only through ratios. The overall scale is a free direction that grows or decays exponentially without the finite-time blow-up of the absolute model.

Introduce the relative abundances $y_i = x_i/m$, which satisfy $\langle y\rangle := \frac{1}{N}\sum_i y_i = 1$ by construction, and let $M = \sum_j x_j$ be the total. The aggregate grows at the population-averaged fitness,

$$g(t) := \frac{d\ln M}{dt} = \frac{1}{M}\sum_j \dot{x}_j = \sum_j \frac{x_j}{M}\,F_j = \frac{1}{N}\sum_j y_j F_j = \langle yF\rangle. \tag{A.3}$$

Subtracting this common scale leaves a closed equation for the relative abundances,

$$\dot{y}_i = y_i\big(F_i - g(t)\big), \qquad F_i = 1 - y_i - (\alpha y)_i, \qquad g(t) = \langle yF\rangle, \tag{A.4}$$

which is a replicator equation, obtained through the standard Lotka–Volterra-to-replicator transformation described by Hofbauer [18]. The aggregate is recovered from $M(t) = M(0)\exp\int_0^t g$, and a positive stationary g is a steadily growing economy.

What the $-g(t)$ term is. The drift $-g(t)$ in (A.4) is the mean-fitness subtraction of the replicator, and it is the term that turns the equation for absolute size into one for relative size. From (A.3), $g(t) = \langle yF\rangle = \sum_j w_j F_j$ is the share-weighted average fitness of the whole economy, with shares $w_j = y_j/N = x_j/M$, and it equals the growth rate of the aggregate, $g(t) = d\ln M/dt$. Every firm feels the same $g(t)$. A firm therefore gains relative size, $\dot{y}_i > 0$, only when its own fitness F_i exceeds the economy-wide average $g(t)$, so $-g(t)$ encodes the requirement to beat the field in order to gain ground. Its role is to enforce the constraint $\langle y\rangle = 1$. One checks from (A.4) that $\frac{d}{dt}\langle y\rangle = \langle yF\rangle - g\langle y\rangle = g\,(1 - \langle y\rangle)$, so $\langle y\rangle = 1$ is preserved, and $g(t) = \langle yF\rangle$ is precisely the value required to do so. Because it is an average over the whole population, $g(t)$ self-averages. Its fluctuations are of order $1/\sqrt{N}$ and vanish in the thermodynamic limit, so it enters the mean-field description as a deterministic common drift rather than as a source of noise.

A.2 The local field and the effective single-site process

We take the high-connectivity limit and treat the regular graph, so that $k_i \equiv C$. Split the interaction felt by node i into its mean and its fluctuation,

$$(\alpha y)_i = -\frac{\mu}{C}\sum_{j\in\partial i} y_j + \frac{\sigma}{\sqrt{C}}\sum_{j\in\partial i} z_{ij}y_j \;\longrightarrow\; -\mu M_1(t) + h_i(t), \tag{A.5}$$

where the neighbourhood average of the mean part tends to the global one, $-\mu M_1$ with $M_1 = \langle y\rangle$, and h_i collects the disorder. (This h_i is the disorder part of the field alone and enters the fitness with a minus sign.) The dynamical cavity argument [9, 15] proceeds by removing node i. The cavity trajectories $y_j^{(i)}$ are independent of the couplings $\{z_{ij}, z_{ji}\}$, and reinstating i perturbs each neighbour by linear response. The field that i exerts on j

has fluctuating part $-\frac{\sigma}{\sqrt{C}} z_{ji} y_i$, so with the single-site response $G_j(t,s) = \delta y_j(t)/\delta\zeta_j(s)$,

$$h_i(t) = \frac{\sigma}{\sqrt{C}} \sum_{j\in\partial i} z_{ij}\, y_j^{(i)}(t) \;-\; \frac{\sigma^2}{C} \sum_{j\in\partial i} z_{ij} z_{ji} \int_0^t ds\, G_j(t,s)\, y_i(s). \tag{A.6}$$

The first sum is a sum of C independent zero-mean terms; by the central limit theorem it becomes a Gaussian noise $\xi_i(t)$ whose covariance is the population autocorrelation,

$$\langle \xi_i(t)\xi_i(s)\rangle = \frac{\sigma^2}{C} \sum_{j\in\partial i} y_j(t) y_j(s) = \sigma^2\, C(t,s), \qquad C(t,s) := \langle y(t) y(s)\rangle. \tag{A.7}$$

The second sum uses $\overline{z_{ij} z_{ji}} = \gamma$ and $\frac{1}{C}\sum_{j\in\partial i} G_j \to G := \langle G_j\rangle$, giving the retarded Onsager reaction $-\gamma\sigma^2 \int_0^t G(t,s)\, y_i(s)\, ds$. Substituting into $F_i = 1 - y_i + \mu M_1 - h_i$ and using (A.4) produces the effective single-site process

$$\boxed{\dot{y} = y\left[1 - y + \mu M_1(t) - g(t) + \sigma\,\eta(t) + \gamma\sigma^2 \int_0^t G(t,s)\, y(s)\, ds\right],} \tag{A.8}$$

with η a unit-strength Gaussian noise of covariance $\langle \eta(t)\eta(s)\rangle = C(t,s)$. Equation (A.8) is the heterogeneous effective process of Aguirre-López [2] with carrying capacity $K = 1$, the competition sign on the mean field, and the two structural additions forced by the scale invariance, namely the common drift $-g(t)$ of Section A.1, and the constraint that the mean abundance is pinned, $M_1 \equiv 1$, rather than left free.

A.3 Self-consistency

The disorder average is described by four functionals of the single-site law of (A.8):

$$M_1(t) = \mathbb{E}[y(t)] = 1, \qquad C(t,s) = \mathbb{E}[y(t)y(s)], \qquad G(t,s) = \mathbb{E}\left[\frac{\delta y(t)}{\delta\zeta(s)}\right]_{\zeta=0}, \tag{A.9}$$

$$g(t) = \mathbb{E}[y(t)F(t)], \qquad F = 1 - y + \mu M_1 + \sigma\eta + \gamma\sigma^2 \int_0^t G\, y. \tag{A.10}$$

The first is the normalization, the middle two are the standard correlation and response that close the noise and the memory, and the last fixes the common drift. The closure for g follows from the normalization: $\dot{M}_1 = \mathbb{E}[yF] - gM_1$, and demanding $\dot{M}_1 = 0$ at $M_1 = 1$ gives $g(t) = \mathbb{E}[yF]$. This has the structure of a solvable random-replicator model, here carrying the GLV self-limitation as well. Solving these self-consistency conditions is a random-replicator mean-field problem related to the effective-process treatments of random Lotka–Volterra systems in Galla [15] and Roy et al. [24]; the assumptions of those models differ from those of the relative model, and the freezing threshold is taken from simulation in the main text.

Appendix B

Finite-size scaling of the exponents across topologies

Chapter 4 reported that the multiscaling ratios are converged in system size while the size–variance exponent β drifts, and gave the drift for the power-law-2.5 graph. Because the drift is the sharpest quantitative caveat on the model, this appendix records the same comparison for every topology of Section 4.4. It separates the finite-size transient from the quantities that have already converged. Figure B.1 and Table B.1 give the size–variance exponent and the highest multiscaling ratio ζ_4/ζ_1 against N, at the fixed dilution $C = N/40$, for each degree distribution. The fully-connected graph is dense and cannot be carried past $N = 16000$ on the available hardware; the regular graph, homogeneous but sparse, provides the homogeneous control at the largest sizes.

The multiscaling ratio (right panel) is flat in N for every topology to within its error above $N \approx 8000$: the homogeneous graphs sit on the fixed-shape value 4 at every size, and each heterogeneous graph holds its own value ($\zeta_4/\zeta_1 \approx 3.4$ for the power-law-3.5 graph, ≈ 3.1 for power-law-2.5, ≈ 2.7 for power-law-2.2) across the range $N = 8000$ to 32000. Carried down to $N = 2000$, the sparse graphs tell the rest of the story: below $N \approx 8000$ their ratios converge toward one another near the fixed-shape value (the power-law-2.5 graph rises from its 3.1 plateau to 3.3 at $N = 4000$ and 3.8 at $N = 2000$, and the regular and power-law-3.5 graphs bunch nearby), so the separation of the plateaus by degree distribution is itself a large-N effect that finite size suppresses. The multiscaling is thus a genuine thermodynamic property of each ensemble, set by the degree distribution and not by the system size, and the empirical value 2.9 is bracketed between the power-law-2.5 and power-law-2.2 graphs at every N where the plateaus have separated.

The size–variance exponent (left panel) behaves differently. It rises with N for every topology across the whole range $N = 2000$ to 32000, but the drift and its limit are organized by the degree distribution. On the homogeneous graphs β is already near 1 and nearly flat: the fully-connected graph sits at ≈ 1.02, the regular graph creeps from 0.73 at $N = 2000$ to 0.99 at 32000 and levels off. The heterogeneous graphs climb further but level off more slowly: the power-law-3.5 graph runs from 0.61 to 0.80 and the power-law-2.5

graph from 0.32 at $N = 2000$ through 0.43 at 8000–16000 to 0.49 at 32000. The step in β per doubling of N shrinks along each curve, the signature of a finite-size approach to a limit rather than an unbounded drift, and extrapolating in $1/N$ (the form that pins the regular graph at $\beta_\infty = 1.00$, $R^2 = 0.999$) sends every topology to a finite value: $\beta_\infty \approx 0.8$ for power-law-3.5 and ≈ 0.5 for power-law-2.5, though for the heavier tail the limit is only loosely constrained by these sizes. The power-law-2.2 graph converges fastest and is already flat over the measured range: 0.16 at $N = 8000$, 0.22 at 16000, 0.23 at 32000, the last two agreeing within their errors, its exponent essentially settled just above the empirical band 0.15–0.20. (It is not carried below $N = 8000$: the most heterogeneous graph is also the most fragile, and at $N = 2000$–4000 most of its realizations freeze, leaving a survivor-biased handful with anomalously high exponents.) Every topology thus appears to approach a finite large-N exponent set by its degree tail; only for the heavy-tailed graphs is that limit still loosely pinned at the sizes reached here. What the sweep establishes is therefore not a single matched exponent but a monotonic control: the degree-tail exponent sets where β lands, spanning the whole range from 1 down through the empirical band, while it leaves the multiscaling ratio fixed. The size-conditioned *shape* is the converged prediction of the model; the absolute exponent is a finite-size quantity whose large-N value the topology tunes toward, and in the most heterogeneous case into the neighbourhood of, the data.

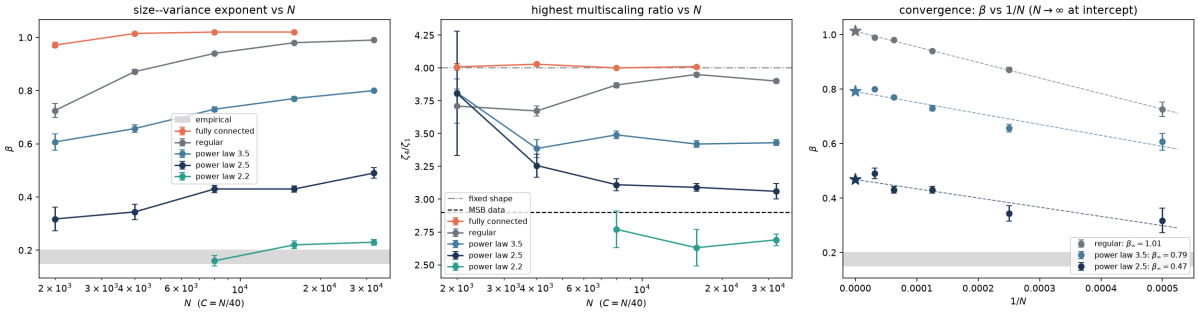


Figure B.1: Finite-size scaling of the exponents at fixed dilution $C = N/40$, for the five degree distributions of Section 4.4 (7–20 economies per point), with the fully-connected, regular, power-law-3.5 and power-law-2.5 curves carried down to $N = 2000$ to show the finite-size approach; the power-law-2.2 graph freezes at these small sizes and is not carried below $N = 8000$. Left: the size–variance exponent β against N; it drifts upward for every topology, but the drift shrinks and the value falls as the degree distribution broadens, from $\beta \approx 1$ on the homogeneous graphs to $\beta \approx 0.23$, converged, on the power-law-2.2 graph, just above the empirical band. Middle: the highest multiscaling ratio ζ_4/ζ_1 against N; flat in N above $N \approx 8000$ for every topology, set by the degree distribution, with the empirical value bracketed between the power-law-2.5 and power-law-2.2 graphs. Right: the same exponents plotted against $1/N$, where the infinite-size limit is the intercept at $1/N = 0$ (stars); the dashed lines are linear fits, which send every graph to a finite β_∞ (the same fit pins the regular graph at 1.00, the power-law-3.5 at 0.79 and the power-law-2.5 at 0.47), evidence that β approaches a finite topology-set limit rather than drifting without bound. The fully-connected graph, dense, is not run beyond $N = 16000$.

		$N = 8000$	$N = 16000$	$N = 32000$
β	fully connected	1.02	1.02	n/a
	regular	0.94	0.98	0.99
	power law 3.5	0.73	0.77	0.80
	power law 2.5	0.43	0.43	0.49
	power law 2.2	0.16	0.22	0.23
ζ_4/ζ_1	fully connected	4.00	4.01	n/a
	regular	3.87	3.95	3.90
	power law 3.5	3.49	3.42	3.43
	power law 2.5	3.11	3.09	3.06
	power law 2.2	2.77	2.63	2.69

Table B.1: The size–variance exponent β and the highest multiscaling ratio ζ_4/ζ_1 against system size, at $C = N/40$, for each degree distribution (standard errors ≤ 0.02 on β and ≤ 0.14 on the ratio; the empirical values are $\beta \approx 0.15$–0.20 and $\zeta_4/\zeta_1 \approx 2.9$). The ratio is flat in N for every topology; β drifts upward but its value is set by the degree distribution, and for the power-law-2.2 graph it has converged just above the empirical band. The ensembles here are independent of those behind the main-text figures; where the same quantity appears in both, the values agree within the quoted errors. The fully-connected graph is dense and is not run at $N = 32000$.

Appendix C

Between-firm growth correlations

Moran, Secchi, and Bouchaud [22] argue that granular models fail because they omit a correlation between firms' growth rates that strengthens with firm size: "the correlation of a firm's growth rate with that of the whole economy and the correlation of the growth rate of firms of similar sizes clearly grow with their size". The model studied here reproduces the size-conditioned statistics through the degree heterogeneity of its interaction graph. The natural question is whether it does so through the same size-growing correlations or through a different channel.

We measure two correlations from the per-firm growth time series $g_i(t) = \Delta \ln S_i$ in the fluctuating state, on the thesis graph (power-law 2.5, mean degree $C = N/40$, $N = 16000$, ten economies). The first is the co-movement of a firm with the aggregate, $\rho_i = \mathrm{corr}_t\big(g_i(t),\, \bar{g}(t)\big)$, where $\bar{g}(t)$ is the cross-sectional mean growth: it measures how strongly a firm tracks the common motion of the economy. The second is the mean pairwise correlation $\mathrm{corr}_t\big(g_i(t),\, g_j(t)\big)$ among firms that fall in the same bin, the direct analogue of Moran, Secchi, and Bouchaud [22]'s "correlation of firms of similar sizes". Each is binned both by firm size and by degree.

Both measures are small and, within the resolved range, nearly flat (Figure C.1). Co-movement with the aggregate is about $\rho \approx 0.09$, with no visible dependence on size or degree. The same-bin pairwise correlation is roughly an order of magnitude smaller, ≈ 0.01, with at most a weak rise to ≈ 0.018 among the largest firms. The correlations identified by Moran, Secchi, and Bouchaud [22] are therefore present but modest, and they do not increase appreciably with either size or degree.

The size-conditioned anomalies therefore arise here *without* the size-growing correlations conjectured by Moran, Secchi, and Bouchaud [22]. Instead, the standard Lotka–Volterra normalization makes the amplitude of a firm's idiosyncratic noise increase with its connectivity (Section 4.4). That is a single-firm mechanism operating against a modest, largely flat background of between-firm correlations. The model is thus an alternative to Moran, Secchi, and Bouchaud [22]'s hypothesis, rather than a realization of it.

The pairwise measure averages over same-bin pairs that are almost all unconnected on the sparse graph, so it may understate correlations between direct trading partners.

The survival mechanism of Section 4.5 also removes the extreme hubs from the fluctuating state: the live-firm degree axis reaches only $k \approx 470$, although the graph tail extends beyond $k \approx 3000$. A direct test on firm-level data with a known interaction network would remove both restrictions.

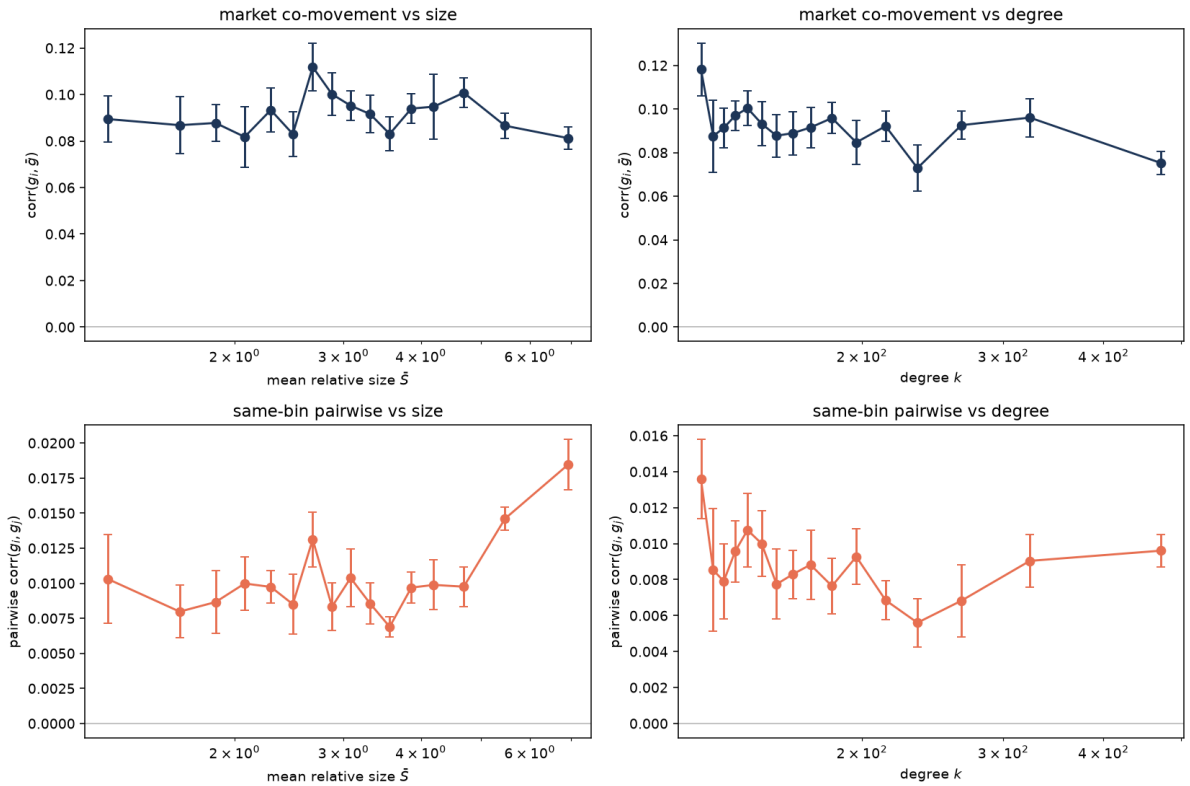


Figure C.1: Between-firm growth correlations in the fluctuating state ($N = 16000$, ten economies, power-law-2.5 graph). Top: the co-movement of each firm with the aggregate, $\mathrm{corr}(g_i, \bar{g})$, against firm size (left) and degree (right); flat at ≈ 0.09 on both axes. Bottom: the mean pairwise correlation among firms in a bin, against size (left) and degree (right); flat at ≈ 0.01, with a weak rise among the largest firms. Neither correlation grows appreciably with size or degree, so the model reproduces the size-conditioned statistics through the degree-dependent noise amplitude rather than through the size-growing correlations of Moran, Secchi, and Bouchaud [22].

Appendix D

Survival and re-entry dynamics

This appendix measures the selection mechanism of Section 4.5 directly. Figure D.1 shows the conditional survival probability for ten independent power-law-2.5 economies of $N = 10000$, with $C = N/40 = 250$, at the operating point of the main text. A firm counts as a survivor when its share remains above 10^{-6} throughout the late window $t \in [180, 240]$. We calculate a rate in each degree bin within each economy and then average the ten rates with equal weight. The bands are 95% t intervals across economies.

Survival falls sharply with degree: the rate is 0.159 ± 0.049 in the lowest resolved bin ($k \approx 75$), 0.044 ± 0.015 at $k \approx 352$, and 0.014 ± 0.008 at $k \approx 985$; no survivor is observed above $k \approx 2760$. This is the direct selection step behind the degree–size relation: connectivity broadens the disorder field but competition removes the typical high-degree firm. The graph construction samples no degrees below $k = 58$, so the figure says nothing about weakly connected firms.

Crossing the threshold is not an absorbing extinction event. In the same ten trajectories we classify a re-entry only when a firm is active, drops below the threshold, and later returns above it; a first activation does not count. Although only 11.3% of firms remain above the threshold throughout the window, 87.5% are active at some point. Among firms that exit at least once, 85.4% re-enter before the window ends (Figure D.2). The re-entry probability conditional on an exit is 0.87 in the lowest resolved degree bin and remains 0.77 at $k \approx 985$, falling only in the sparsely populated extreme tail. The low persistent-survival probability therefore records turnover around the threshold rather than irreversible firm loss. Immigration continually reseeds low shares, while the fluctuating interaction field can return a firm to the active set.

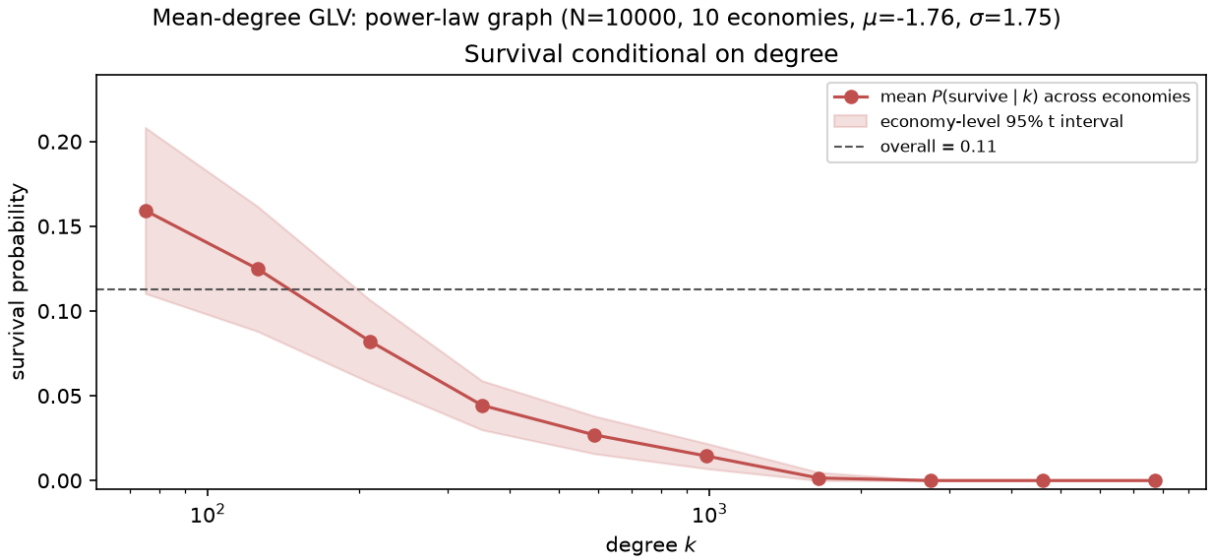


Figure D.1: Survival conditional on degree in the mean-degree power-law ensemble ($N = 10000$, $C = 250$, ten economies, $\mu = -1.76$, $\sigma = 1.75$). Each point is the unweighted mean of the within-economy survival probabilities; shading is the 95% interval across economies. A firm is counted as surviving only if its share stays above 10^{-6} throughout $t \in [180, 240]$. The steady fall across the resolved support is the competitive selection of hubs, measured directly.

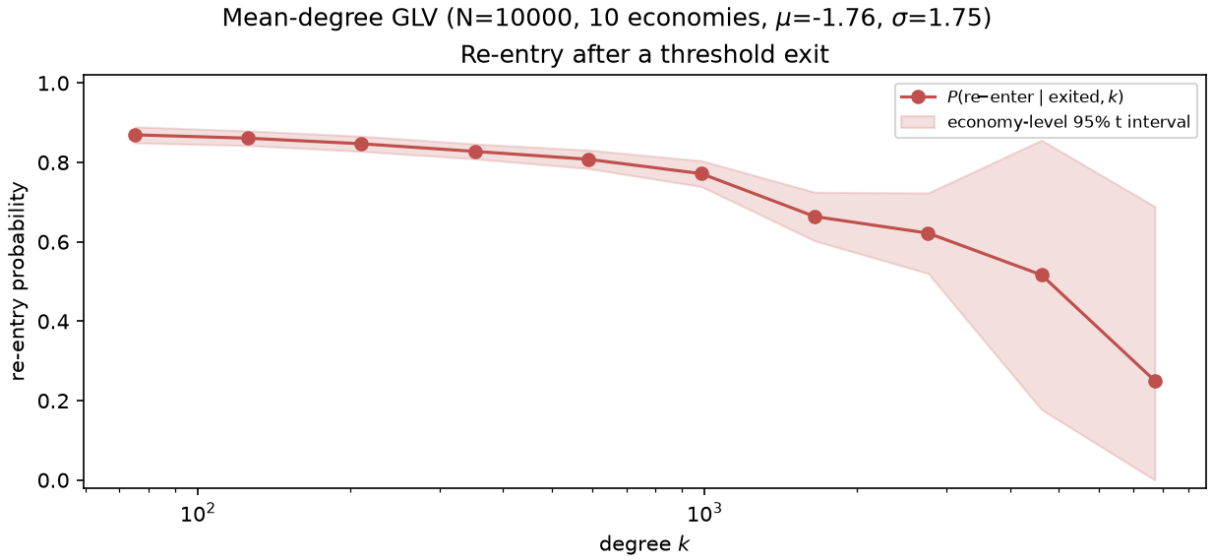


Figure D.2: Re-entry after a threshold exit in the mean-degree power-law ensemble ($N = 10000$, $C = 250$, ten economies, $\mu = -1.76$, $\sigma = 1.75$). A re-entry requires an active–inactive–active sequence within $t \in [180, 240]$; first activation is excluded. Each point is the unweighted mean of the within-economy probability $P(\text{re–enter} \mid \text{exited}, k)$, with 95% intervals across economies. Most exits are followed by a return, so crossing the threshold is rarely an absorbing extinction.

Appendix E

Firm-size distribution

Figure E.1 measures the stationary distribution of relative sizes $S_i = Nw_i$ at the operating point: ten independent $N = 10000$ economies, each observed over the late window $t \in [180, 240]$. The left panel keeps every observation; its lower plateau is set by the immigration floor. The reporting cut $S = 0.1$, applied only in the upper-tail panel, removes this floor-dominated region and is not a graph cutoff. On log–log axes the active-firm tail bends sharply instead of following a straight Pareto segment. A fit of the complementary-tail exponent in each economy, from its upper 2% of active observations, gives $\hat{\mu} = 4.40$ with a 95% t interval $[4.02, 4.78]$; a Zipf tail has exponent near one. The upper tail at this operating point is far thinner, effectively truncated.

The main results of the thesis are therefore claims about the size-conditioned growth fluctuations and their network origin, not about the cross-sectional size distribution, which the model at this operating point does not reproduce.

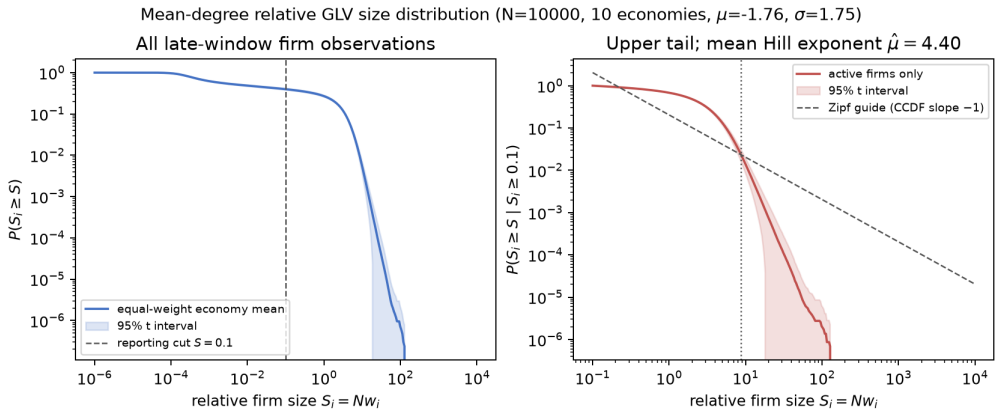


Figure E.1: Stationary relative firm-size distribution in the mean-degree power-law ensemble ($N = 10000$, $C = 250$, ten economies, $\mu = -1.76$, $\sigma = 1.75$). Curves are equal-weight means of economy-level complementary distribution functions; shading is the 95% interval across economies. Left: all late-window observations. Right: conditional upper tail of active firms, with strong curvature and a complementary-tail exponent near 4.4 against the Zipf benchmark near 1.

Appendix F

Numerical methods and measurement protocol

F.1 Integration

The relative model is integrated in physical time, with the composition kept in bounded variables and the aggregate scale carried in its logarithm. This avoids numerical overflow despite exponential aggregate growth. Internally the solver evolves the shares $w_i = S_i/N$, which live on the probability simplex and are renormalized onto it at each output time; we integrate (3.6) in this form together with $\mathrm{d}\ln m/\mathrm{d}t = g_{\mathrm{eff}}$ with an explicit Runge–Kutta solver throughout. At the system sizes reached in the text (N up to 3.2×10^4) the implicit solvers we tried switched to a dense Jacobian and became prohibitively slow, whereas the explicit method integrates one run to $t = 250$ in about ten seconds at $N = 10^4$.

F.2 Volatility estimator

The per-firm growth-rate volatility is measured with the mean-absolute-deviation estimator

$$\sigma_i = \sqrt{\tfrac{\pi}{2}}\,\big\langle\, \big|\, g_i - \langle g_i \rangle \,\big|\, \big\rangle, \tag{F.1}$$

rescaled by the Gaussian-consistent factor $\sqrt{\pi/2}$ and used in place of the standard deviation for robustness against the fat tails of the growth rates.

F.3 Size–volatility fit

The immigration floor makes the smallest firms quiet, producing a plateau at the bottom of the size range. We therefore fit $\sigma(\bar{S})$ only along the monotone decline above the plateau and report the coefficient of determination R^2, rather than impose one slope on a curve that is not a power law throughout. The exponent quoted in the text, $\beta = 0.45 \pm 0.06$, is

the median and economy-to-economy standard deviation of this decline-branch slope over twenty independent disorder realizations at $N = 16000$; the spread is set by the disorder realization rather than the fit, whose pooled-decline R^2 is 0.99. All exponents, including the higher moments ζ_q, are fitted per economy and averaged (Section 4.2); pooled fits are quoted only for comparison.

F.4 Robustness of the measurement protocol

The protocol of Chapter 4 contains two arbitrary numerical choices: growth rates are read over the fixed horizon $\Delta t = 0.5$, and the trajectories are integrated at relative tolerance 10^{-4}. This section varies both on the graph of the main text (power law 2.5, $C = N/40$) at $N = 8000$, ten economies, and re-measures the three headline statistics: the size–volatility exponent β, the quenched moment ratios ζ_q/ζ_1, and the excess kurtosis of the growth-rate distribution.

For the horizon, each economy is integrated once and stored at fine output resolution, and every Δt from 0.25 to 4 is measured on the same trajectory, so any difference between the columns of Figure F.1 is an estimator effect and not a resampling of the dynamics. The moment ratios do not move: over a factor of sixteen in Δt they stay at 1.86, 2.55, 3.1 with every shift inside one standard error. The exponent is flat within the economy-to-economy spread up to $\Delta t = 1$ ($\beta = 0.40 \pm 0.07$ at $\Delta t = 0.25$ against 0.41 ± 0.07 at the production value) and then climbs steadily, 0.49 ± 0.05 at $\Delta t = 2$ to 0.58 ± 0.06 at $\Delta t = 4$: once Δt exceeds the correlation time of the chaotic state, the endpoints of an increment are two nearly independent draws from the firm's stationary distribution, so the measured variance no longer grows with the horizon and reflects the width of that distribution rather than the local diffusion. The correlation time is measured on the same trajectories: the firm-averaged autocorrelation of $\ln S$ is 0.82 at lag 1, 0.54 at lag 2 and 0.18 at lag 4, a half-life of 2.3 ± 0.3 time units, so the flat range $\Delta t \leq 1$ sits below half the correlation time and the drift sets in where $C(\Delta t)$ decays. The horizon is therefore part of the definition of β, as it is in the data, where the exponent is reported at a yearly horizon. The ratios ζ_q/ζ_1 show no comparable dependence. Identifying the production horizon $\Delta t = 0.5$ with one year places the firm memory time at four to six years; every horizon below that time returns the same exponent. The excess kurtosis falls from 9.2 to 4.2 over the same range as successive increments aggregate toward the Gaussian; the tent stays above the Laplace reference at every lag.

For the tolerance, the first three economies are re-integrated at relative tolerance 10^{-6} (absolute 10^{-9}) and measured at the production horizon. Individual trajectories in a chaotic state cannot coincide across tolerances; the statistics do. The per-economy exponents move from 0.59, 0.47, 0.41 to 0.58, 0.48, 0.45, a largest shift of 0.04 against the ± 0.07 spread between economies, and the moment ratios and kurtosis shift within their standard errors.

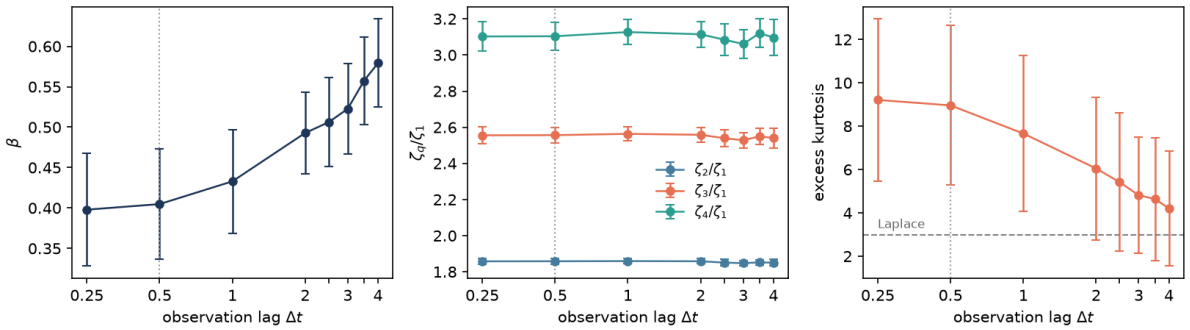


Figure F.1: Dependence of the headline statistics on the observation lag Δt, ten economies of $N = 8000$ ($C = 200$) at the operating point of the main text, all lags measured on the same trajectories. Left: the size–volatility exponent β (median and economy-to-economy standard deviation). Centre: the quenched moment ratios ζ_q/ζ_1 (mean and standard error). Right: excess kurtosis of the growth-rate distribution (median and spread), with the Laplace reference at 3. The dotted line marks the production value $\Delta t = 0.5$.

F.5 Code availability

The model, the measurement protocols described above, and a generator for every reproducible figure in this thesis are available at `https://github.com/NotLudovico/dynamical-model-for-economics`. The repository includes the dynamical mean-field solver of Appendix A and its validation against direct simulation.